

Massachusetts Institute of Technology
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CALIPSO: A Test Configuration for an Aluminum-Fueled Buoyancy Engine

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Executive Summary

A small, compact buoyancy engine powered by aluminum fuel rather than solely on battery power will enable unmanned underwater vehicles to operate for longer periods of time. This provides significant improvements in efficiency and also saves on operating costs. Aluminum fuel specifically is notably energy dense, but requires technical development to prove its reaction with water (or ammonia) can drive reliable and consistent power and gas generation for buoyancy change. This team has developed CALIPSO, a test configuration which serves as a proof of concept for changing buoyancy through an aluminum fuel and water (or ammonia) reaction.

CALIPSO makes use of three primary material segments: one for fuel, one for deionized water, and one reaction bladder. A system of four pumps, tubing, and electronic control together drive and combine the ingredients from the fuel and deionized water and combine them into the reaction bladder to enable this reaction. Two independent waterproof enclosures house different halves of the system: the dry electronics, and the flooded chamber housing the ingredients and bladder. The reaction occurs in the flooded chamber, producing gas, from which the bladder can inflate or deflate, expelling or taking in outside water, and in turn impose the system's buoyancy change. The system as a whole, for testing, is held in tension and kept stationary with a tension scale used to measure an effective change in buoyancy from the system.

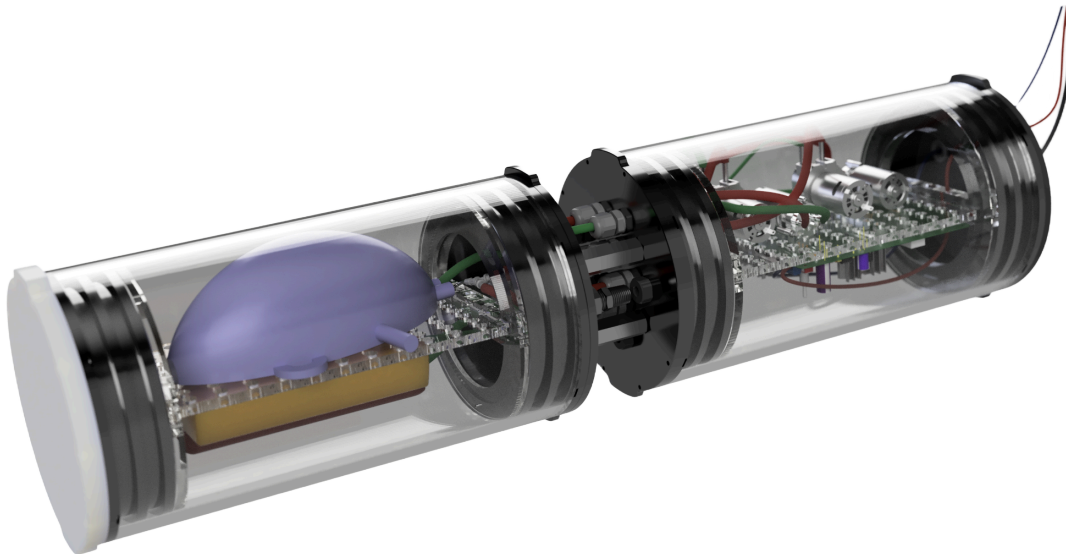


Figure 1. Render of system concept.

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1. Introduction & Background

1.1 Context and Motivation

Ocean monitoring is critical for understanding the effects of climate change, identifying valuable resources underwater, and learning about ecosystems that support our survival [1]. Unmanned Underwater Vehicles (UUVs) are frequently deployed for ocean monitoring because they can access deep and shallow waters to collect oceanographic data [1]. The most common UUVs today are propeller-powered Autonomous Underwater Vehicles (AUVs) and gliders that utilize buoyancy engines to ride ocean currents. Propellers help AUVs steer and move faster than the currents. Meanwhile, buoyancy engines change a glider's net buoyancy for ascent and descent. The glider can move slowly with ocean currents for months at a time and stop at a particular depth for exploration. Despite differences in propulsion systems, both have tradeoffs in efficiency.

A major limitation of all UUVs is their dependence on battery power. Traditional lithium-ion batteries only support 3 days of AUV operation at 3 knots, failing to meet the demands of long underwater missions that can last up to several months [2]. While gliders can ride ocean currents at a particular depth for months at a time, changing buoyancy states consumes significant energy, and frequent depth changes may not be supported by the glider's battery supply [3]. Overall, UUVs have insufficient operational ranges because they depend on batteries. Costly surface ships (~\$70,000 per day) are deployed to monitor UUVs, but this drastically reduces their lifetime value. For UUVs to adequately support our ocean monitoring needs, we need to increase their operational range while eliminating dependence on limited-energy batteries and expensive surface ships.

During last semester, the team proposed a robust and cost-effective buoyancy engine, named PUMPY, that runs on aluminum fuel, which boasts 42x greater energy density than traditional Li-Ion batteries [2]. By successfully harnessing aluminum's high energy density, it can enhance the operational range of UUVs while reducing reliance on battery power. PUMPY initiates a chemical reaction between aluminum fuel and an aqueous ammonia solution to produce compressible gasses. These gases are utilized to adjust the buoyancy of a UUV at depths up to 500m for extended durations of time. By leveraging pressure and heat energy from these gasses, PUMPY captures 25% of aluminum's chemical energy and reduces battery consumption by 90%. PUMPY is also 10x more affordable than existing buoyancy engines. Overall, PUMPY advances ocean monitoring by enhancing the operational range of UUVs, providing an exciting cost-and-energy-efficient solution. This semester, CALIPSO, as a descoped proof-of-concept for PUMPY, is developed to prove aluminum fuel reaction's capability to induce buoyancy change underwater.

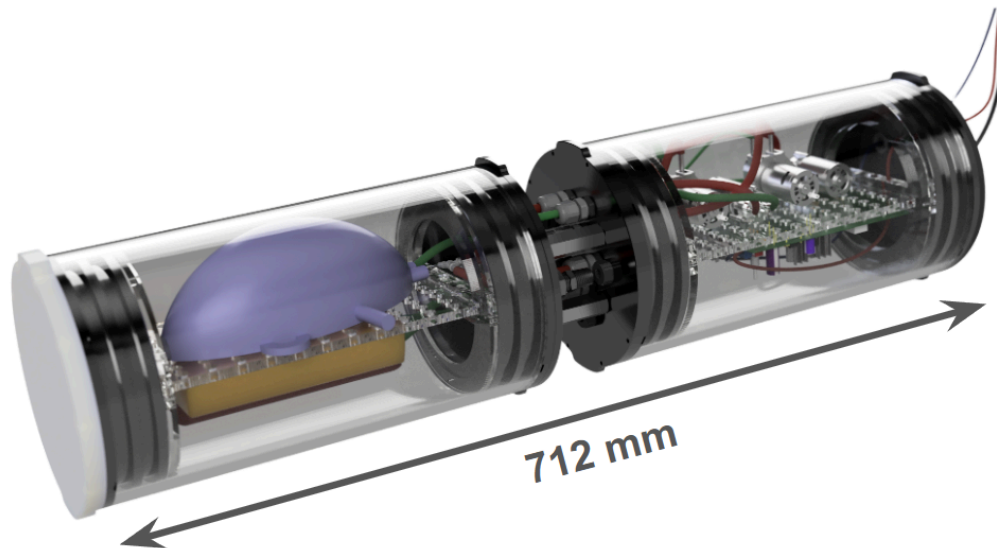


Figure 1.1 CAD Model of Aluminum Fuel Buoyancy Engine Proof-of-Concept, CALIPSO

1.2 Descoping

During the last design cycle, a buoyancy engine, PUMPY, runs on aluminum fuel which is 42 times more energy dense than lithium batteries was designed to address the high cost and limited range problem of current UUVs.

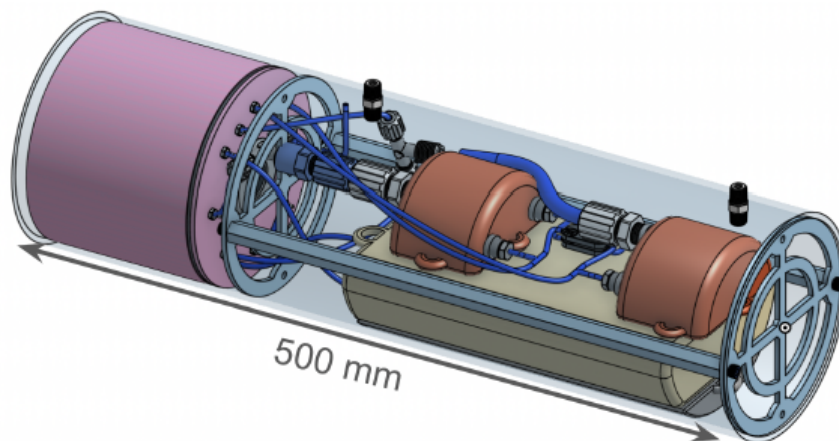


Figure 1.2 CAD Model of Previous Buoyancy Engine Concept, PUMPY

Before being able to pursue the full scope buoyancy engine shown above, we had to prove that such a system could even work. The team needs to prove that a buoyancy engine could utilize this reaction successfully to induce a buoyancy change. To prove this main function, it comes with challenges inherent to the nature of the system.

The ammonia and aluminum reaction can be dangerous and requires special care to handle the reactants. The 500 m depth requires specialized waterproof enclosures or pressure vessels to handle extreme underwater conditions, as well as low temperature and high pressure cracking pumps to perform necessary operations. Additionally, features like fine buoyancy control and moment balance poses unnecessary hardwares which increase system complexity. In

all, these issues only compound at great depths would make the system dangerous, high cost, hard to test, and very difficult to rapidly prototype, which is key in the proof-of-concept stage the team are in.

In order to de-risk and de-scope changes were made to the functional requirements when transitioning to making CALIPSO. These include a reduced operational depth, pressure and temperature range. Since they were not essential to proving the functionality of the system and increased its complexity, fine buoyancy control and moment control were removed as requirements for the test setup. Importantly, the vehicle compatibility and the amount of water needed to be displaced by the system to create a buoyancy change are kept the same for both as proof of its scalability. The de-risking process also involves a change in the reaction. Since ammonia is dangerous to handle, it is substituted with water. This water reaction yields less gasses as there is no ammonia gas generated in the reaction. This means that more fuel was needed to achieve the same gas yields as with ammonia. The system requirements were modified so that a full test of the contained, underwater reaction could be set-up, run, and deconstructed within about 3 hours. These changes were also made so that a typical pool could serve as a sufficient testing environment.

Table 1.1: High-level functional requirements comparison for PUMPY and CALIPSO

	Requirement	Description	PUMPY	CALIPSO
1	Operational Depth	Depth to target for dive & return	500 m	5m
2	Operational Pressure Range	Pressure range of operation	1 - 50 atm	1 - 1.5 atm
3	Operational Temperature Range	Temperature range of operation	0 - 49° C	15 - 49° C
4	Fine Buoyancy Control	Precise control over depth shifts	Yes	No
5	Moment Balance	Limit shifts in CG relative to CB	Yes	No
6	Buoyancy Displacement	Min. volume of water that engine can displace	≥ 350 mL	≥ 350 mL

7	Vehicle Compatibility	Max. diameter for engine to be compatible with target vehicles	< 0.15 m	< 0.15 m
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As a follow up to the test configuration, several key changes would be necessary for performance in the real ocean environment and depth. Hardware rated to withstand greater pressure and a higher stored reactant volume would allow sustained dives at depth. A removal of the current tether for onboard power supply and a decrease in design volume, and/or consolidation into a single chamber would be needed for compatibility with typical UUVs. The only change that is not intuitively scalable is the chemical reaction. The reaction in the current test configuration used to inflate the bladder is that of aluminum and water. Compared to water, ammonia has a lower enthalpy of vaporization, will expand with less energy, and condense much more slowly, all of which would be more favorable for repeated reactions especially at depth. In the context of the testing, which aims to verify if the core mechanisms of this engine can function reliably, these benefits are not worth the extra handling caution needed for ammonia. While the water and aluminum reaction will be used for prototyping, we highlight that this ammonia material change will be important for reliable low depth operation.

1.3 Overview of Al-fueled Buoyancy Capabilities

Generally, buoyancy engines are devices that displace mass or volume to change the buoyancy of a vehicle. CALIPSO displaces water volume to change the net buoyancy of a UUV. As depicted in Figure 1.3, CALIPSO would initiate a chemical reaction between aluminum and water which produces hydrogen, aluminum byproducts, and a significant amount of heat which is shown in equation 1. The heat is then used to vaporize the ammonia creating a significant amount of gas. The gas-to-water ratio within the engine can be changed by creating or expelling gas, resulting in net positive buoyancy ($F_b > F_g$) for ascent, net negative buoyancy ($F_b < F_g$) for descent, or neutral buoyancy ($F_b = F_g$).

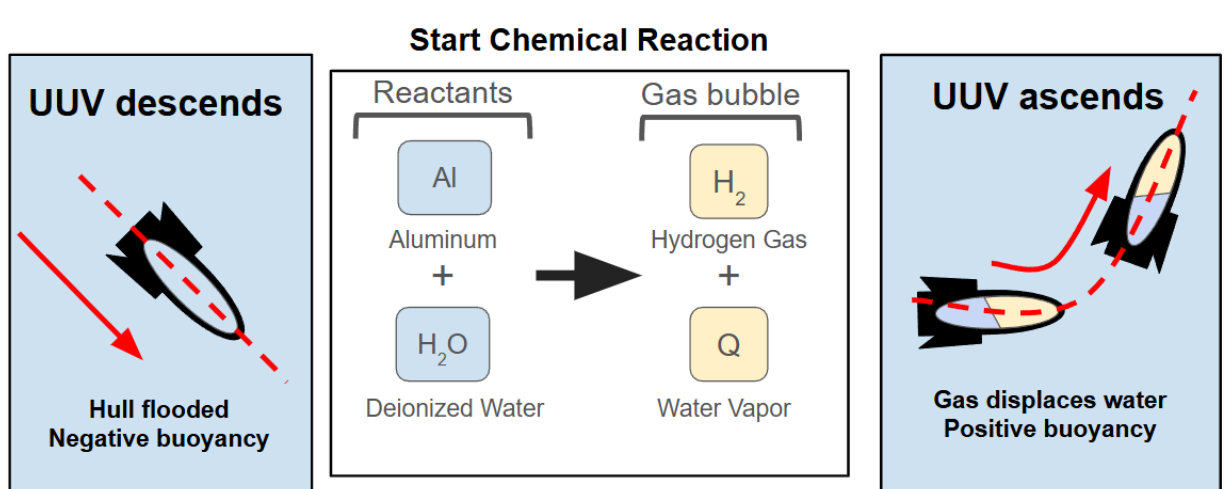
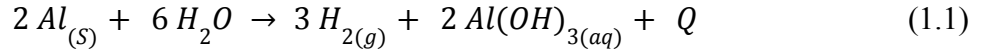


Figure 1.3: Diagram of how CALIPSO utilizes a chemical reaction to increase the gas-to-water ratio inside UUVs. This displaces water volume to change net buoyancy, allowing transition from descent to ascent.



With the ability to adjust the gas-to-water ratio, in the previous semester, PUMPY provided UUVs with four major capabilities: ascent, descent, fine buoyancy control, and moment balancing. Figure 1.4 shows that a UUV using PUMPY has a dive profile similar to gliders. It can travel up to a depth of 500 meters and ascends/descends at a pitch angle set by deployers. Furthermore, if a UUV wants to monitor something, it can fine-tune its buoyancy to become neutrally buoyant and stop at a specific depth. Then, the UUV can utilize propellers for further exploration along that depth. Lastly, since a UUV frequently transitions between different pitch angles throughout its dive cycle, there are relative shifts between the vehicle's center of gravity and center of buoyancy. If there are significant shifts between center of buoyancy and center of gravity in the direction of tilt, nose diving may occur. PUMPY can adjust the vehicle's pitch angle to mitigate this risk.

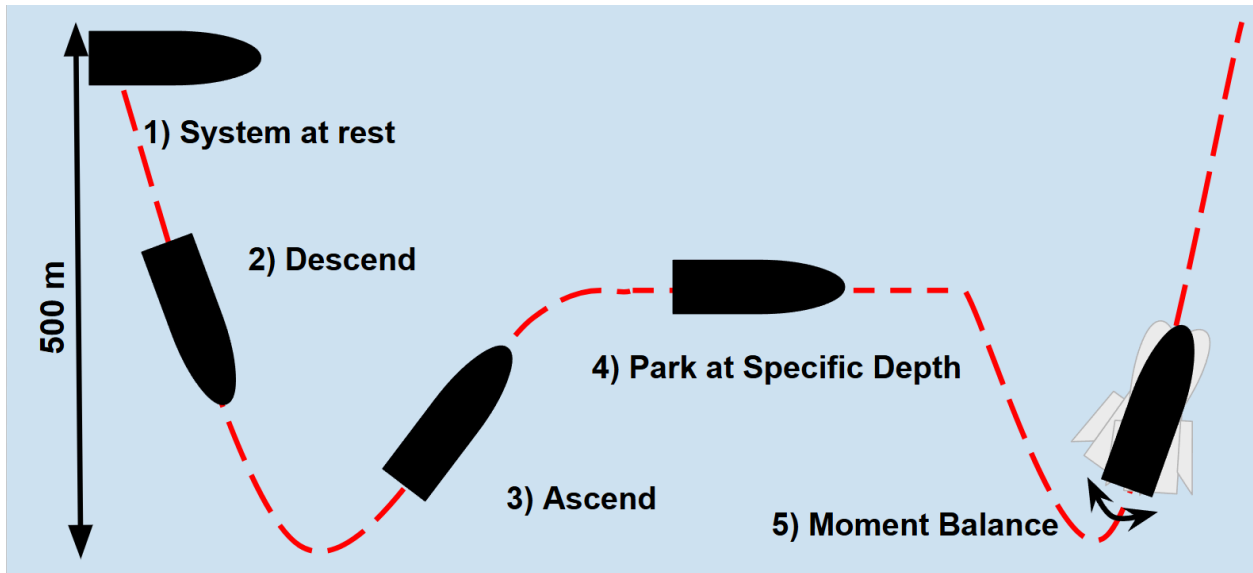


Figure 1.4: Dive profile of a UUV using PUMPY. The UUV can ascend, descend, stop at a particular depth when the buoyancy engine adjusts for neutral buoyancy, and adjust pitch angle via moment balancing.

As mentioned earlier, to descope the system, moment balance and fine buoyancy control is removed for CALIPSO. It now has 2 major capabilities: ascent and descent. Figure 1.5 shows the updated dive profile. Importantly, CALIPSO will also prove that the system is able to perform multiple cycles of ascent and descent.

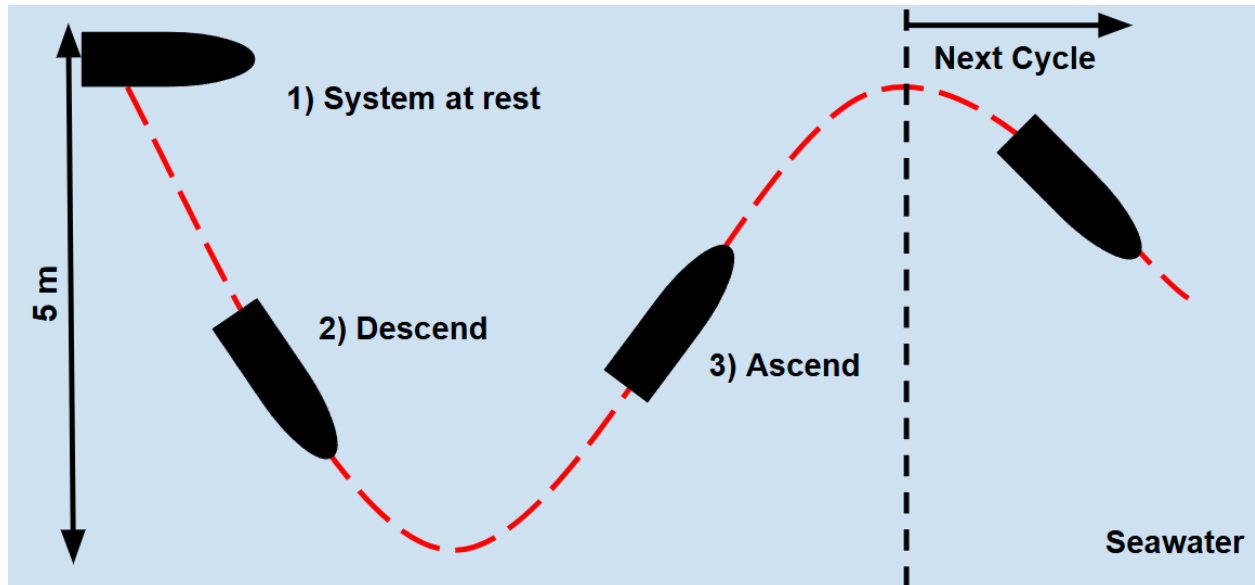


Figure 1.5: Dive profile of CALIPSO. The testing configuration can ascend and descend for multiple cycles.

1.4 System Design Overview

In this section, we provide an overview of CALIPSO's system design, and its two actions: ascent and descent.

1.4.1 Ascent

CALIPSO displaces water to change the buoyant volume of a UUV in order to initiate ascent. The engine operates within a flooded section of the UUV, where deionized water and aluminum paste are stored. These reactants are dispensed into a metal y-connector using pumps. Gases produced from the reaction then inflate a gas bladder in the flooded hull. As the bladder expands, it pushes water out of the flooded section. We initially achieve the desired buoyant volume from these gases when the reaction occurs at high heat. However, the hot reaction gases will eventually cool as heat is lost to the cold ocean. If any water is allowed to enter the flooded hull during or after cooling, the bladder will deflate and lose buoyancy, interfering with ascent. CALIPSO prevents this through a check valve on the flooded hull, which ensures unidirectional flow. The valve allows water to flow out when the reaction occurs, but prevents water from entering the flooded section and compressing the bladder during ascent. Note that the reactant pumps, as well as the controllers and electronic components required for them cannot sit in seawater, so we've configured an additional electronic enclosure for them. The component layout diagram for ascent is shown in Figure 1.6.

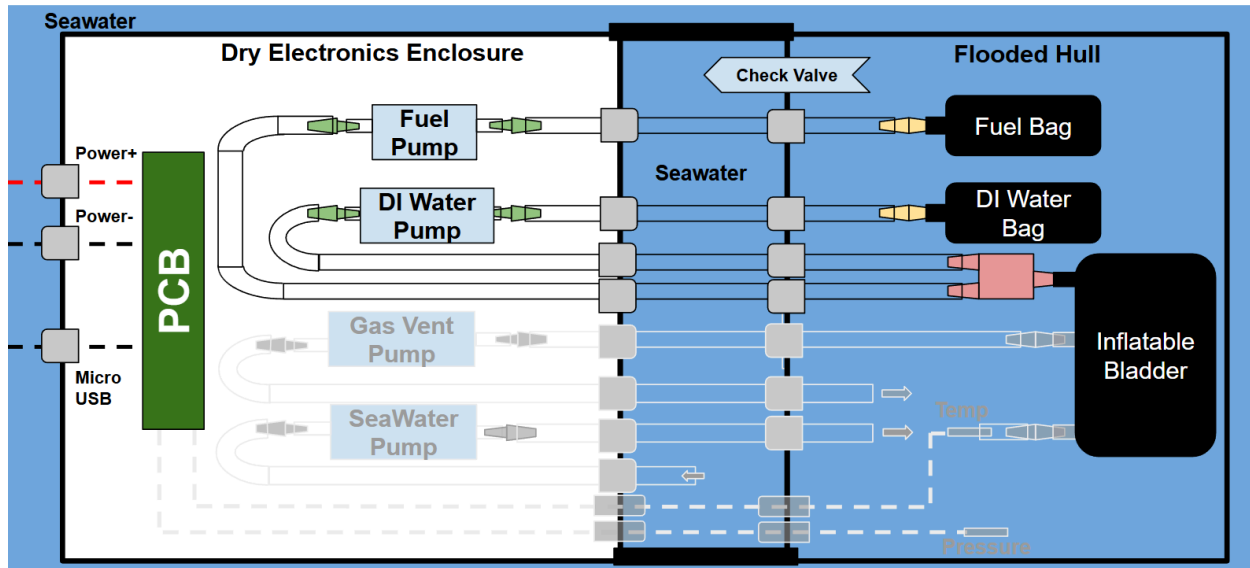


Figure 1.6: Block diagram of key features supporting ascent capabilities. Deionized water and aluminum fuels are combined to produce gases that inflate a gas bladder, increasing buoyant volume.

1.4.2 Descent

For the UUV to descend, CALIPSO now needs to reduce buoyant volume. To accomplish this the system takes advantage of a two part gas venting system to eliminate the need to pump against a very large pressure differential, like a normal buoyancy engine did. In the first part of this process a gas venting pump moves the gas products from the inflatable bladder into the flooded hull. During this step the pump operates against no pressure differential. The inflatable bladder ensures that the pressure within the bladder equals the pressure within the flooded hull as it can expand and contract to maintain equilibrium. This means the gas products are simply transferred from within the bladder walls to outside of them while remaining in the same pressure environment. The second step of this process utilizes the pressure of the external seawater to negate any large pressure differential between the flooded hull and external environment. During this step the sea water pump pulls in sea water from outside the hull and pumps it into the flooded hull. This pump builds up the required 2 psi pressure differential between flooded hull and external environment to reach the cracking pressure of the check valve, which once activated allows the gas to flow through it and vent into the sea water. This pump runs until all the gas within the flooded hull is vented through the check valve. The use of the check valve and dual pump setup means this entire operation regardless of depth will always have a maximum pressure differential to generate of 2 psi. The replacement of gas volume within the flooded hull with sea water returns the test configuration to a state of negative buoyancy and allows for it to descend. The component layout diagram for descent is shown in Figure 1.7.

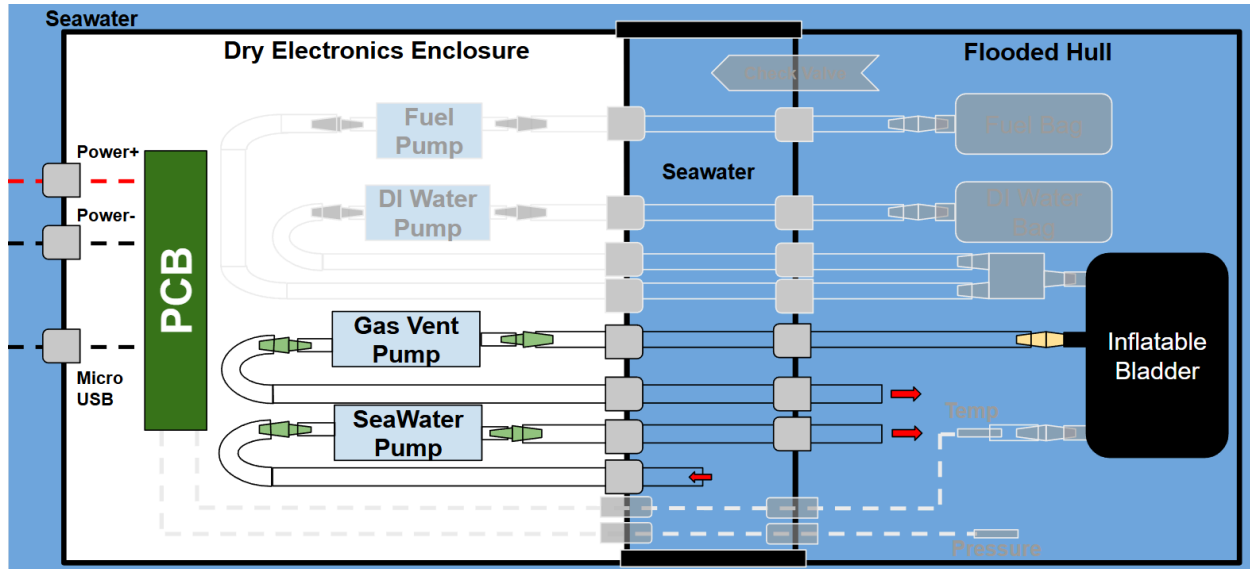


Figure 1.7: Block diagram of seawater pump and gas venting pump supporting descent capabilities. A gas pump vents gas from the bladder while a seawater pump increases internal pressure to push gas bubbles out of the flooded section.

1.4.3 Integrated System

To summarize, when the engine is used to initiate ascent, deionized water and aluminum pumps in the electronic enclosure draw fuel from their respective bags in the flooded hull. These reactants combine to initiate a chemical reaction, which produces gas products that inflate the bladder whose volume affects buoyancy. Additionally, the bladder has a gas venting pump, which displaces gas from the bladder to the flooded hull. These gases are released through a check valve at the ocean-hull interface, which activates under a pressure differential. A seawater draws in water to increase hull pressure for adjusting bladder volume. It is also critical for increasing vehicle density to initiate descent. All electronic components are protected from water in a separate enclosure. The overall layout diagram for CALIPSO is shown in Figure 1.8.

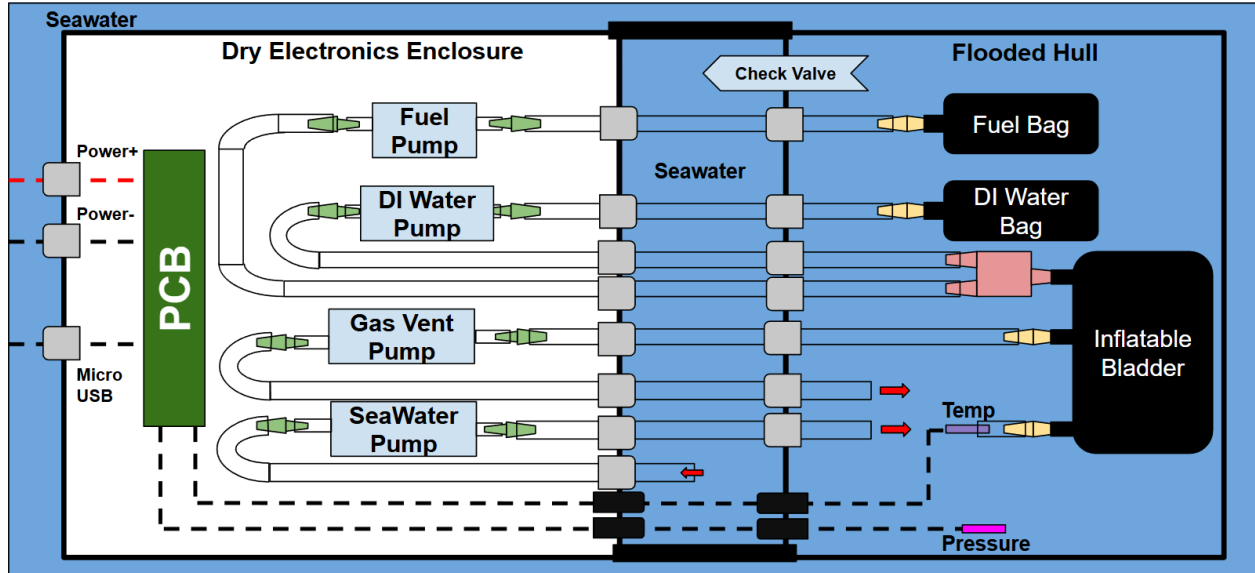


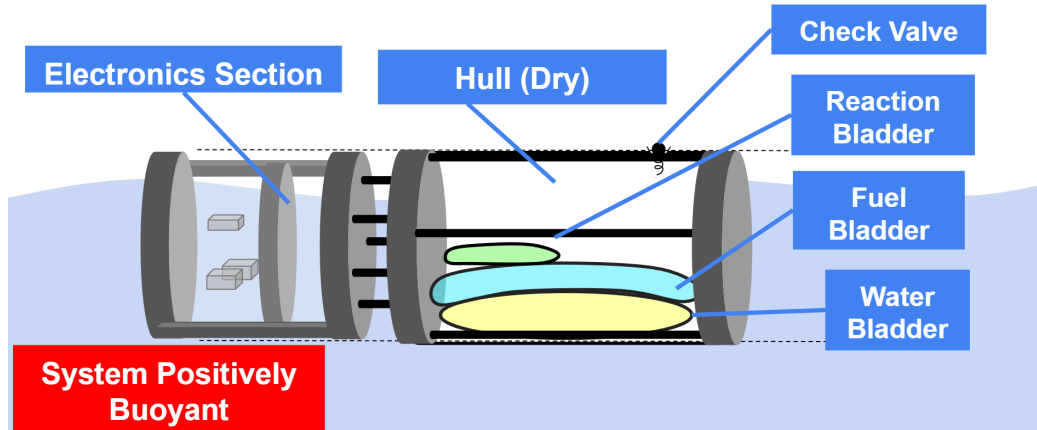
Figure 1.8: High-level block diagram of CALIPSO. Shows detailed mapping of fuel displacement, reaction routing, bladder venting, tubing routing, and sensor placements.

In addition to the system layout, CALIPSO's design and use case follow specific operating conditions. As depicted by the diagrams in Table 1.2, before the system is first deployed, the flooded hull is filled with air. The system then initiates descent by drawing in water to replace the less dense air, which causes the system to sink. Prior to reaching a maximum depth of 5 meters, the system initiates a reaction between aluminum paste and deionized water, which generates gases that inflate bladders. The increase in bladder volume increases the buoyant force on the system until it ascends. Once the system reaches the surface again, the gases in the bladders are vented for deflation, allowing the system to descend again. Table 1.2 describes the pressure, temperature, and volume conditions at each stage of the dive.

Table 1.2: Map of pressure, temperature, and volume throughout CALIPSO dive cycle.

Stage	Description				
1	System at Rest • Dry hull is leveled at the surface by deployers for a neutral moment. • Bladder is empty before first dive <table><tr><td>$P_{ocean} = 1 \text{ atm}$</td><td>$P_{hull} = 1 \text{ atm}$</td><td>$V_{bladder} = 0 \text{ mL}$</td><td>$T = 25^{\circ}\text{C}$</td></tr></table>	$P_{ocean} = 1 \text{ atm}$	$P_{hull} = 1 \text{ atm}$	$V_{bladder} = 0 \text{ mL}$	$T = 25^{\circ}\text{C}$
$P_{ocean} = 1 \text{ atm}$	$P_{hull} = 1 \text{ atm}$	$V_{bladder} = 0 \text{ mL}$	$T = 25^{\circ}\text{C}$		

Step 1 System at Rest



2

Descent

- Draw in sea water to initiate descent
- Ocean pressure increases with depth h , $P_{ocean} = P_{atm} + \rho_{ocean}gh$
- Temperature decreases along thermocline

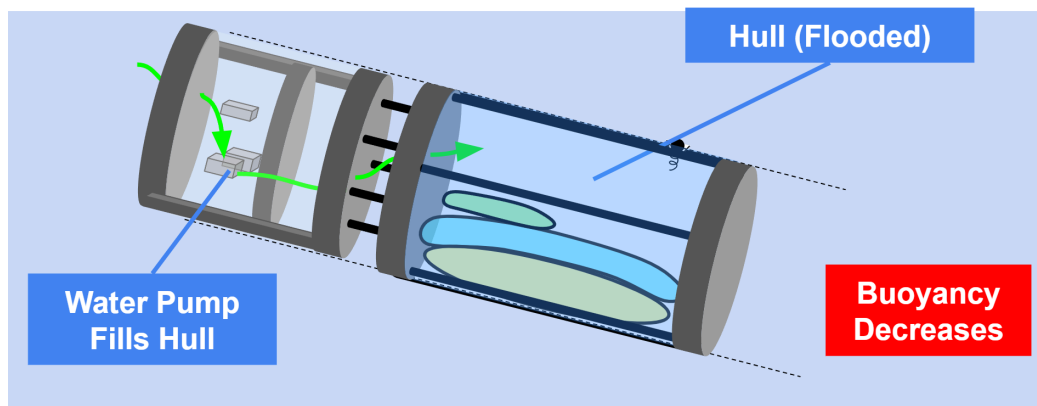
$$P_{ocean} \rightarrow 1.5 \text{ atm}$$

$$P_{hull} = 1 \text{ atm}$$

$$V_{bladder} = 0 \text{ mL}$$

$$T \rightarrow 15^\circ\text{C}$$

Step 2 Descending



3

Reaction

- P_{hull} and $P_{bladder}$ both start at 1 atm and are always equal to each other via hydrostatic equilibrium
- Reaction generates gas to raise $P_{bladder}$ to ~ 1.5 atm, displacing internal water and inflating the bladder
- Reaction raises temperature of bladder and tubing to 200°C while ocean temperature remains unchanged

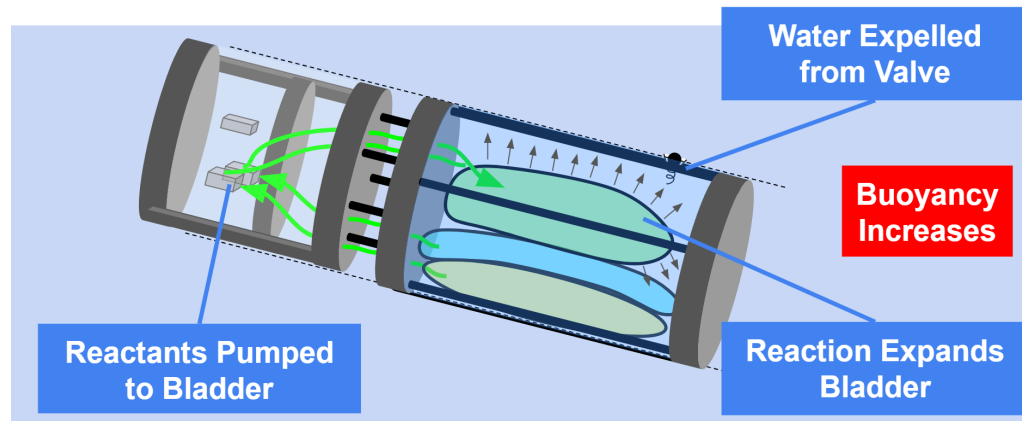
$$P_{ocean} \sim 1.5 \text{ atm}$$

$$P_{hull} \sim 1.65 \text{ atm}$$

$$V_{bladder} \rightarrow 350 \text{ mL}$$

$$T \sim 15^{\circ}\text{C}$$

Step 3 Ascent (Reaction Start)



4

Transition to descent

- Bladder volume decrease as gas venting out from bladder to hull
- Gas filling the top of the flooded hull

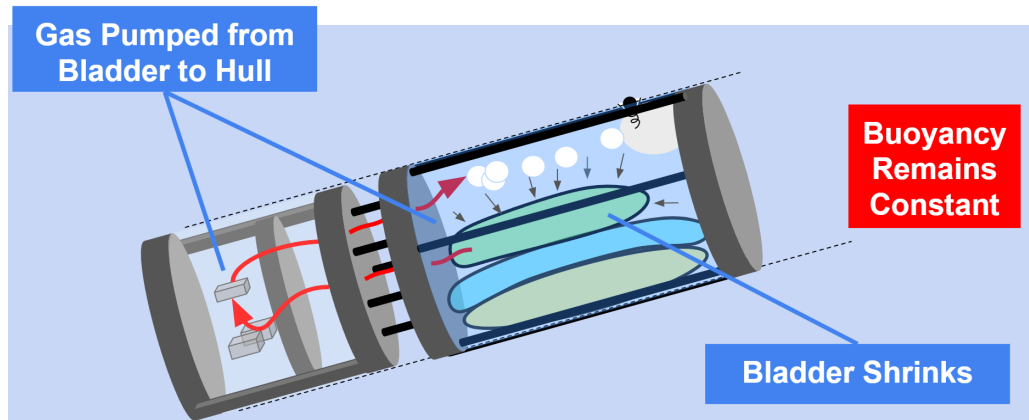
$$P_{ocean} \sim 1.5 \text{ atm}$$

$$P_{hull} \sim 1.65 \text{ atm}$$

$$V_{bladder} \rightarrow 0 \text{ mL}$$

$$T \sim 15^{\circ}\text{C}$$

Step 4 Bladder Vented



5

Decent (again)

- Pressure decreases approaching surface
- Ocean temperature increases along ascent thermocline
- Seawater pump draws in water from ocean to the flooded hull, which activates check valve, allowing gas to escape from hull to ocean, thus decrease buoyancy
- Repeat Step 2 - 5 as operator desired

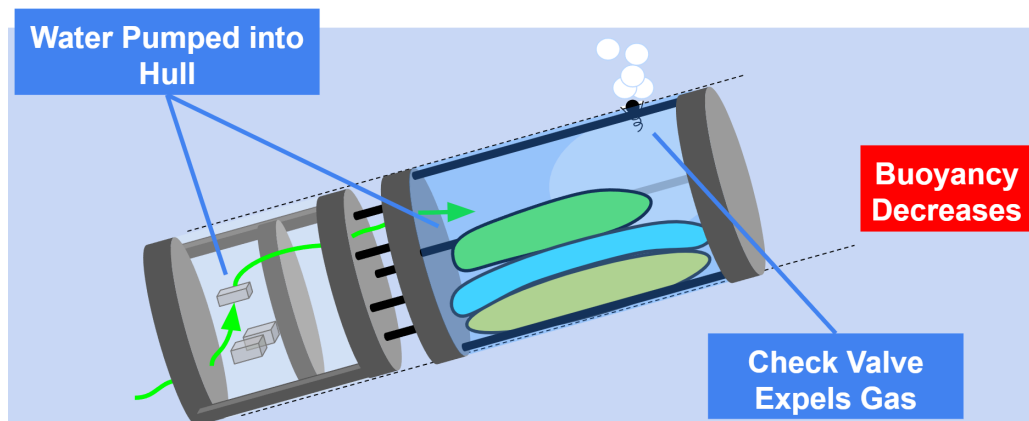
$$P_{ocean} \sim 1 \text{ atm}$$

$$P_{hull} \sim 1.15 \text{ atm}$$

$$V_{bladder} \sim 0 \text{ mL}$$

$$T \sim 25^{\circ}\text{C}$$

Step 5 Ascent Finishes



1.5 Buoyancy Dynamics, Mass, & Volume

In order to understand how buoyancy is changed throughout the dive cycle, we consider a free body diagram of our system, as shown in Figure 1.9. For this, we assume that the engine is placed in a UUV that is neutrally buoyant and experiences no moment from the UUV, such that we can consider our engine alone. In the diagram, F_B represents the buoyant force on the vehicle, F_G represents the gravitational force, and F_D represents the drag force.

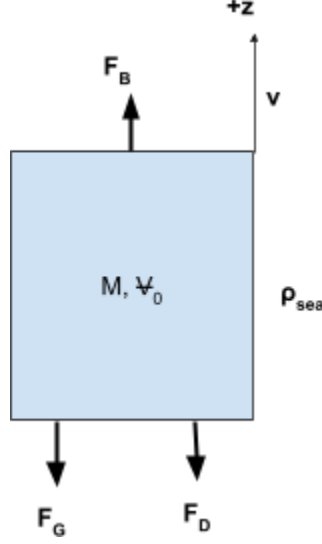


Figure 1.9: Free body diagram of vertical forces acting on UUV while traveling in the ocean. M and V_0 represent mass and fixed volume of the vehicle.

The forces on the engine are defined as follows:

$$F_G = -Mg \quad (1.2)$$

$$F_D = -\frac{1}{2}\rho_{sea}C_DAv|v| \quad (1.3)$$

$$F_B = V_0\rho_{sea}g + V_B(t)\rho_{sea}g \quad (1.4)$$

In these equations, ρ_{sea} is the density of seawater, g is the gravitational acceleration, C_D is the drag coefficient, A is the characteristic area of the vehicle, and v is the velocity of the vehicle at a given point in time. The drag force always opposes the direction of motion of the vehicle. Here, it is drawn as if the vehicle was in its ascending state. M and V_0 represent the fixed mass and fixed volume of the vehicle, which do not change throughout the dive cycle because the engine retains waste products to conserve mass. $V_B(t)$ represents the additional buoyant volume

from the bladders, which changes as the bladders are inflated, and can therefore be thought of as a function of the flow into the bladder. Thus, for some flow Q of gas into the bladders, equation (1.4) becomes:

$$F_B = \forall_0 \rho_{sea} g + \rho_{sea} g \int_0^t Q(\tau) d\tau \quad (1.5)$$

where t represents time after flow is initiated, until the bladders reach their maximum buoyant volume of 350 mL. The net force on the engine is then given as:

$$F_{engine} = F_B - F_D - F_G \quad (1.6)$$

This equation is core to the design of an effective buoyancy engine and determines the relationship between fixed mass and volume that must be maintained in order to ensure the desired buoyancy conditions. Assuming drag may be neglected as the vehicle is moving slowly, when the gravitational terms exceed the buoyant force, ($F_G > F_B$), the net force is negative ($F_{engine} < 0$, “negatively buoyant”) and the vehicle sinks. Similarly, when the buoyant force exceeds the gravitational force ($F_G < F_B$), the net force is positive ($F_{engine} > 0$, “positively buoyant”), and the vehicle floats or ascends upwards. When the two are equal, the net force is zero, and the vehicle is considered neutrally buoyant, and does not experience any acceleration. These equations can be used to develop the required relationship between mass and volume, helping us set the mass and volume budgets for our engine.

To do so, we first considered the desired additional volume $\forall_B(t)$ corresponding to each of the three buoyancy states, which were set by our sponsor’s requirements.

Negative buoyancy: $\forall_B(t) = 0$ ml

Neutral buoyancy: $\forall_B(t) = 150$ ml

Positive buoyancy: $\forall_B(t) = 350$ ml

Using this, we then tracked expected volume for each component during our design process to find our V_0 , which was ultimately estimated at 5380 mL. This allowed us to determine the constraints on allowable mass range from our negative and positive buoyancy conditions, as well as our target mass from our desired neutral buoyancy condition, as detailed in Table 1.3.

Table 1.3: Mass target for engine

Condition	F_B (expression)	F_B (N)	Mass Constraint (kg)
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Negative Buoyancy, $V_B(t) = 0$ ml	$V_0 \rho_{sea} g$	54.4	> 5.54
Positive Buoyancy, $V_B(t) = 350$ ml	$V_0 \rho_{sea} g + V_B(t) \rho_{sea} g$	57.9	< 5.85
Neutral Buoyancy, $V_B(t) = 150$ ml	$V_0 \rho_{sea} g + V_B(t) \rho_{sea} g$	56	5.7

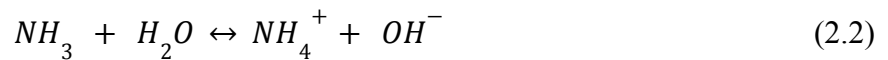
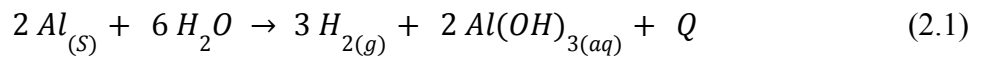
The desired mass for our system is 5.7 kg, with an acceptable range of 5.54 to 5.7 kg. Our design meets our mass budget constraints at 3.65 kg, with room to additionally tune using weights to meet our neutral buoyancy target. CALIPSO did not fulfill the mass design objective so further weight reduction will be needed to produce more buoyancy.

2. Aluminum-Fueled Chemical Reaction

2.1 Fuel Reaction

CALIPSO leverages the products of the exothermic reaction between aluminum and water to displace volume and generate a desired buoyancy. As seen in equation 2.1, the products of this reaction are hydrogen gas, aluminum byproduct, and a significant amount of energy. Therefore, to fully utilize the reaction, an ammonia solution is introduced to the system since the excess heat can be used to vaporize the ammonia and generate additional buoyancy.

Now we'll look a little closer at our reactants. In order to achieve this reaction spontaneously in an underwater vehicle, the aluminum must first be treated and processed. Aluminum, like many metals, oxidizes when introduced to oxygen and forms a surface layer that blocks the aluminum from reaction with water. To overcome this, the aluminum is treated with gallium and indium. Next, the aluminum needs to be ground down and placed in an oil suspension to prepare it to be delivered via pumps in an underwater vehicle. The final result is a fuel paste that consists of 60% aluminum by mass and 40% oil. For the other reactants, due to ammonia's tendency to evaporate, a 20% aqueous ammonia solution by weight supplies both the water and ammonia that is utilized in the reaction. Equation 2.2 is the stoichiometric equation for aqueous ammonia [4].



For our experiments and the following derivations, the ammonia solution described is actually just pure DI water with 20% of it being devoted to being vaporized (like the ammonia was) and 80% being used in the reaction. This is due to safety concerns of handling ammonia.

Later in the scaling up section, we will prove how the viability of a system using DI water can prove the viability of a system using ammonia solution.

Table 2.1: Parameter assumptions for ideal gas law

Ideal Gas Law Parameter	Value	Description
Pressure (P)	1.5 atm	Maximum pressure that will exist in the flooded chamber
Volume (V)	350 mL	Volume needed to generate positive buoyancy
Gas Constant (R)	8.31 J/(mol-K)	Both hydrogen and steam are ideal gases
Temperature (T)	384.15 K	The boiling point of water at 1.5 atm

To find the exact volume of reaction needed to generate the volume of gas necessary to meet our buoyancy calculations, more exact temperature modeling and experimentation would be necessary. However, by making some assumptions and considering worst case scenarios, minimum values can be found to ensure positive buoyancy is achieved. Using values listed in Table 2.1 as well as the ideal gas law, we can calculate the moles required to inflate the bladder.

$$n = \frac{PV}{RT} = 1.66 * 10^{-2} \text{ mol} \quad (2.3)$$

Then, utilizing stoichiometry, the moles of reactants and products were solved. Then, using molar masses, the mass of each was calculated. Lastly, using density the volume of each value was found. These calculations result in the following volume needs:[5]

$$V_{fuel} = 0.211 \text{ mL} \quad (2.4)$$

$$V_{Ammonia\ solution} = 0.500 \text{ mL} \quad (2.5)$$

2.2 Heat Released & Condensation (Johan Thesis)

Previously, we've established that the engine works by generating a bubble of hydrogen and steam gasses after reacting aluminum with D.I.. In this section, we describe the total amount of heat released by the reaction and the total energy needed to vaporize the targeted quantity of D.I. water . Additionally, working with heat within a relatively cold testing environment introduces complications with condensation, which must be accounted for. To estimate heat loss,

a steady state thermal resistor network was assumed for the bladder surrounded by 15°C container water.

The amount of heat released by the reaction is 15.6 kJ/g of aluminum fuel, assuming a 100% yield. A detailed explanation of this value is given in the appendix section A.1. Then, the reaction heat for 0.21 g of aluminum fuel is:

$$Q_{rxn} = m_{Al} \cdot 15.6 [kJ] = 3.1 [kJ] \quad (2.6)$$

The reaction heat is distributed to the products of the reaction and vaporizes pure ammonia (recall there is 40 wt% oil in the paste as well):

$$Q_{rxn} = (mC_p \Delta T)_{H_2} + (mC_p \Delta T)_{Al(OH)_3} + (mC_p \Delta T)_{oil} + [m(C_p \Delta T + H_{vap})]_{H_2O} \quad (2.7)$$

By using the properties of reactants and products in Table 2.2, we find that the total heat required to vaporize water, Q_{rxn} , is 0.89kJ.

Table 2.2. Properties of reactants and products in 1.5 atm condition.

M_{Al}	26.98	g/mol
M_{H_2}	2.02	g/mol
$M_{Al(OH)_3}$	78.00	g/mol
$C_{p Al(OH)_3}$	1.2	J/(gK)
$C_{p H_2}$	14.3	J/(gK)
$C_{p oil}$	2.0	J/(gK)
$H_{vap H_2O}$	2227	J/g
∇	350	ml
ΔT	96	K

Overall, the heat released by the reaction is around 3 times larger than the heat used to vaporize water and inflate the bladder. This reveals that there is a large remainder of heat being wasted. We could add excess water to capture more reaction heat for more water vapor production, decreasing the amount of fuel needed.

Most importantly, we need to account for the rate of heat loss the engine will face over time. The bladder is surrounded by ambient seawater at a temperature close to 0°C, so condensation is expected to deflate the bladder as soon as water and ammonia droplets lose enough energy. We use a heat loss model, shown in Figure 2.2, considering a steady state

scenario with conduction across a perfectly spherical bladder. Hence, flow associated with heat loss is negligible.

$$Q = \frac{\Delta T}{R_{TOT}} \quad (2.8)$$

$$\Delta T = T_w - T_g \quad (2.9)$$

$$R_{TOT} = \frac{R_2 - R_1}{4\pi R_1 R_2 k_r} \quad (2.10)$$

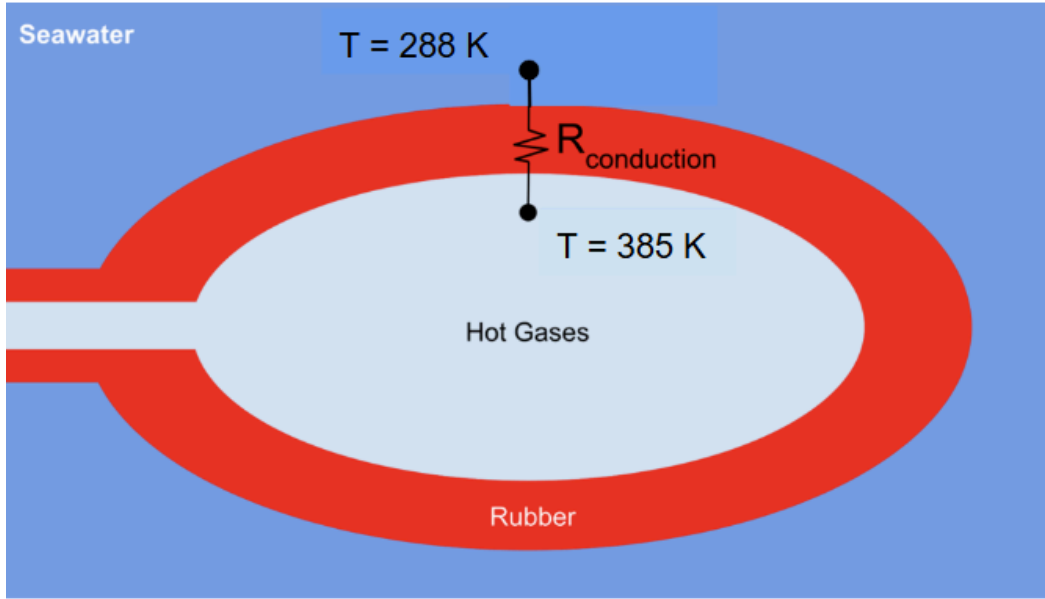


Figure 2.2: Thermal resistor diagram showing the heat loss from hot reaction gases to colder external ocean. The model assumes a steady state, spherical bladder system.

Overall, the gasses inside the bladder experience significant heat loss (Q) and condensation ($\bar{m}$). Thus, it is important for the engine design to capture the gases quickly and avoid further buoyancy loss.

$$Q = -127 \text{ W}$$

$$\bar{m} = \frac{Q}{H_{vap}} = \frac{-127 [W]}{2227 [J/g]} \cdot 60 = -0.0034 \text{ g/min} \quad (2.11)$$

Note: Assumption of 100% yields may come down to around 30% in order to capture the fastest portion of the reaction in the shortest time span.

3. Design Description

3.1 System Overview & Requirements

Controlling flow precisely between components, and isolating the rest to ensure no unexpected mixing, is of primary concern with system design.

3.2 Reactants

3.2.1 Reactant Storage

Storing of the reactants presents a challenge primarily with maintaining a zero pressure gradient across the reactant storage container walls. A solid container would necessitate moving parts to alleviate the negative pressure generated when reactants are pumped out, such as a syringe setup. Although this configuration was initially pursued given the predicted high viscosity of the fuel and associated high required pressure differential to convey it, once the viscosity was confirmed to be in a pumpable range, flexible storage bladders were instead chosen as the ideal option.

Chemical compatibility of the bladders within this test configuration is simpler to solve than in the proposed full scope design due to the substitution of distilled water instead of high concentration aqueous ammonia. Based on this substitution silicone was chosen for the reactant storage material due to the ease of sourcing and cost effectiveness, while still maintaining chemical compatibility and allowing for successful testing to prove the concepts of an aluminum fuelled buoyancy engine. Silicone blood pressure cuff bladders were selected due to high reliability and integrated connection ports for barb fittings. Additionally, they handle expansion well for large volumes of reactants providing a high quality and cost effective alternative to custom reactant storage manufacturing.

3.2.2 Aluminum Fuel Characterization

Solving the issue of conveying the aluminum fuel at consistent and precise rates required the usage of the manufactured aluminum fuel paste as opposed to the more commonly used prepared aluminum pellets. As discussed in the fuel reaction section above this Aluminum paste is a mixture of finely ground aluminum particles between 20 and 60 microns in average Particle Size (APS), canola oil, and fumed silica. The two additives in this solution are to improve the flow characteristics of the fuel suspension. Suspensions are heterogeneous mixtures with insoluble particles that will eventually settle out of suspension over time due to their size being larger than in a true solution such as a colloid [6]. A true colloid will have particle sizes ranging from 1 to 1000 nanometers which do not settle out of the solution. The addition of the fumed silica serves two purposes to ease the difficulties associated with the use of this suspension, to act as a thickener and a shear thinning additive. While these seem to oppose each other the thickening aspect increases the viscosity of the fluid to retain the suspended particles in solution longer reducing the issues associated with them settling and the shear thinning property improves

the flow characteristics of the fuel allows it to be more conducive to pumping throughout this test configuration.

Designing the reactant system for this test configuration meant first characterizing the fluid properties of the fuel suspension to determine the correct hardware requirements for the system. However the extremely reactive nature of the aluminum fuel necessitated selecting a viscosity corollary to conduct hands on testing. Stainless steel powder was ultimately selected since it is widely available in similar powder sizes, cost effective, relatively nontoxic in its alloy state, and one of the least reactive metals even in its powder form. Stainless steel compared to other metal powders is not classified as explosive under dust explosibility testing standard E1226–12a based on its high activation energy even when dispersed in oxygen and relatively low energy yield [7]. The selected stainless steel particle size of 15 microns was notably higher than the observed aluminum particle size although size analysis could not be conducted on the final fuel particles within suspension due to reactivity concerns. The aluminum particles were prepared separately from this design cycle by another team and near the end of the cycle, following the viscosity testing, although this did not prevent our selected hardware from functioning properly with the fuel. The selected stainless steel powder with an APS of 15 microns was mixed with 50 cSt silicone bath fluids and fumed silica, and amounts were chosen to match the amount of particles in solution between the actual fuel and the stainless steel stand-in. This relationship was chosen as the basis for the correlation since suspensions derive their flow characteristics, much of their viscosity variability, and abrasiveness from the size and distribution of the particles. More specifically, Einstein's equation, given in equation 3.1, proves that viscosity of a suspension of spherical particles only relies on the viscosity of the suspension medium (μ_0), and volume of particles in suspension (V)

$$\mu = \mu_0(1 + 2.5 V) \quad (3.1)$$

Using the stainless steel suspension viscosity testing was conducted using the viscosity cup test method outlined in ASTM 1200 and 4212 although a syringe vessel without a plunger was used as the test vessel instead of the called for Ford or Dip-Type Viscosity Cup. Water was used as the known viscosity substance and the time it took for 40 mL of water to flow from the vessel was measured. Then the vessel was filled with 40 mL of the stainless steel suspension and again the time was recorded. Three trials were conducted and the unknown dynamic viscosity of the stainless steel suspension was calculated based on the relationship shown below in Equation 3.2 and 3.3. Using this equation the stainless steel suspension, and by extension of the viscosity correlation the aluminum fuel suspension, has an estimated viscosity of 30.9 to 48.7 cSt and a dynamic viscosity of 100.0 to 157.5 cP. These numbers are reasonable given the silicone oil kinematic viscosity of 50 cSt. Then these characteristics were used to derive our system requirements based on well known fluid dynamics equations.

$$\frac{\eta}{\rho} \sim kt, \quad k - \text{mathematical constant} \quad (3.2)$$

$$\frac{\eta_w \rho_f t_f}{\rho_w t_w} = \eta_f, \text{ where } f \rightarrow \text{unknown fluid} \& w \rightarrow \text{water} \quad (3.3)$$

3.2.3 Reactant Pumps

3.2.3.1 Aluminum Fuel Pump

The system requirements needed for the reactant system are primarily the optimal tubing internal diameter and the pressure head required for the reactant pumps. Thus using the above calculated estimated viscosities, the required pressure curve in Figure 3.1 was calculated based on Equation 3.3 Hagen Poiseuille's equation for laminar flow of a viscous fluid through a long thin pipe.

$$Q = \frac{\pi P r^4}{8 \eta L} \quad (3.3)$$

Q Volumetric Flow [m/s] η Dynamic Viscosity [$N \cdot s/m$]

P Pressure Differential [Pa] L Pipe Length [m]

r Internal Radius [m]

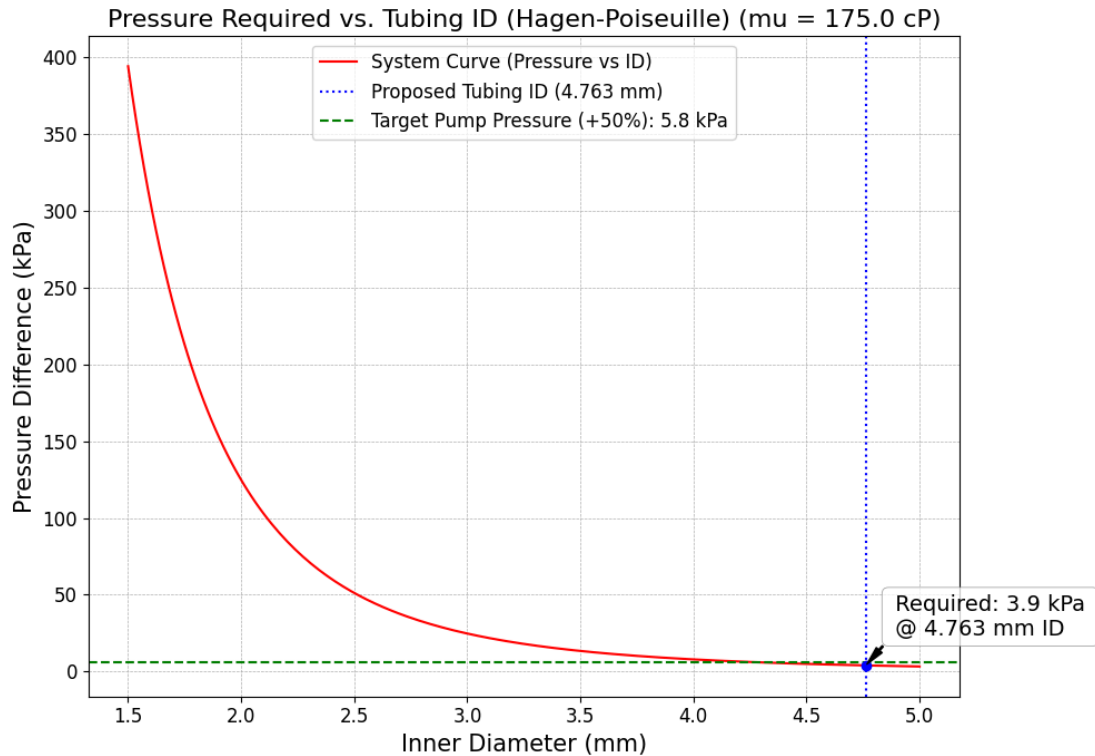


Figure 3.1: A graph of the required pump head pressure to successfully convey the aluminum fuel mixture through the reactant system based on a pipe length of one meter, a dynamic viscosity of 175 centipoise, and a selected internal tube diameter of 4.763 mm or 3/16”.

As seen in the figure the actual required pressure head is 3.9 kPa when using the rounded upper range of dynamic viscosity of 175 centipoise. This reflects only what can barely convey the fluid however so an additional 50% was added as a safety factor to account for differences between the estimate and actual viscosity as well as any inclined sections of tubing that may increase the required head pressure. The tubing diameter was selected to minimize the required head pressure, ensure a standardized size, and still achieve an acceptable bend radius of 9/16”. Using the equation and this graphed relationship the required head pressure of the aluminum fuel pump is 5.8 kPa.

Peristaltic pumps were selected as the reactant pumps for their reliable dosage amounts as well as their lack of contamination risk and fantastic chemical compatibility due to the liquid never leaving the tubing. The functioning of this pump class is based on the principle of peristalsis in which the rollers compress the tubing forcing fluid forwards, then when the tube reopens the leading end is still sealed by the rollers causing a negative pressure region that sucks the fluid into the empty section before it is again compressed by the next rollers. These rotating rollers create continuous pressure waves that continuously convey fluid which is visualized in the cross section shown in Figure 3.2 and the selected pump in its disassembled state in Figure 3.3. Although these pumps generate less pressure head than other pump varieties they can easily supply the necessary pressure head to reliably convey the fuel based on the 5.8 kPa requirement. The specific pump selected for the aluminum fuel was a Gikfun AE1207 12V Peristaltic Dosing Pump with specs shown below in Table 3.1.

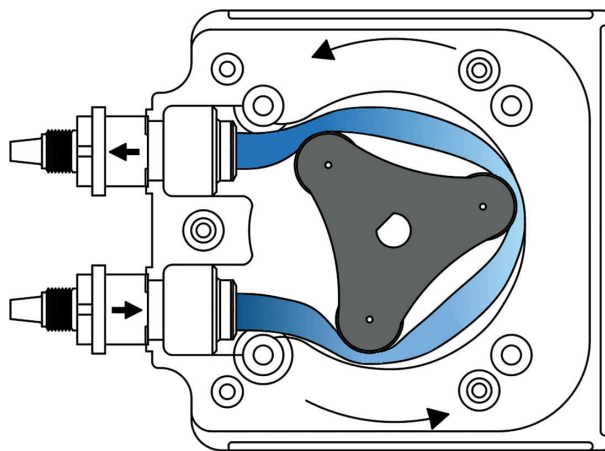


Figure 3.2: The cross section of a peristaltic pump which visualizes the principle of peristalsis.

Table 3.1: A table of the specifications for the selected peristaltic dosing pump.

	Dosing Pumps
Dimensions	63 x 45 x 40 mm
Mass	95 g
Voltage	12 V
Max Pressure	21 KPa
Temperature	0 - 40 °C
Flow Rate	0 - 100 mL/min
Cost	\$12



Figure 3.3: The selected pump in its disassembled state, displaying the inner workings of the chosen pump.

This pump was chosen on the basis of cost effectiveness given it met all other criteria required. With a maximum flow rate of 100 mL/min with water this should have no issue achieving the maximum dosage rate required for the reactants. The operational temperature range fits within the system requirements and it will be operating in the dry electronics enclosure so temperature extremes are not expected. The compact form factor and light mass of 95 grams will allow for easier integration of these pumps into the enclosure. Finally with a max head pressure of 21 KPa, these pumps are comfortably above the required pressure to reliably convey the aluminum fuel confirming these pumps will serve the test configuration well.

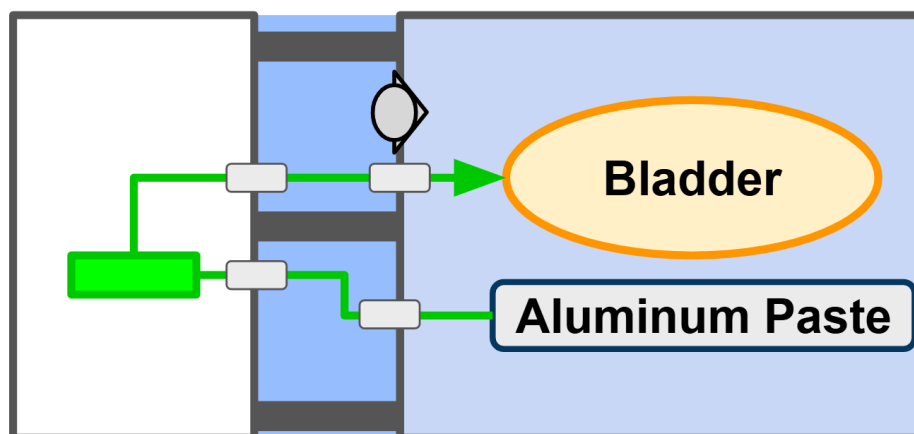


Figure 3.4: A flow diagram for the aluminum fuel pump in the reactant system.

3.2.3.2 Distilled Water Pump

The second reactant pump is the distilled water pump which must reliably convey distilled water from the reactant bag to the reaction chamber at a consistent rate. While this pump will be conveying a different fluid, the same Gikfun 12V dosing pump was selected for its cost effectiveness and the other positives discussed above. The figure below shows the operational block diagram for the distilled water pump. When operating the Distilled water pump conveys water from the reactant bag to the reaction bladder against no pressure differential.

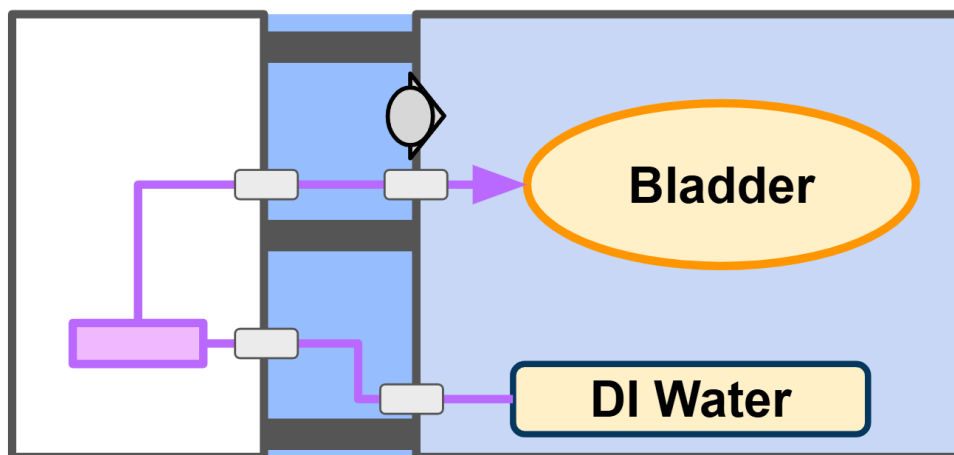


Figure 3.5: A flow diagram for the distilled water pump in the reactant system.

3.2.4 Pump Testing

After pump selection the next step is to determine the estimated time each pump needs to run to dispense the required amount of reactant to generate the gas volume needed to fill the bladder and have positive buoyancy within the test configuration. To determine these rates the pumps were tested at the minimum duty cycle that allowed for steady operation which was 35% or 4.2 Volts. Figure 3.6 below shows the results of this testing with distilled water and the stainless steel suspension. The pumps were not tested with the aluminum fuel suspension due to the dangers discussed previously, instead the estimated rates for aluminum fuel were determined based on the density conversion from the stainless steel suspension to the aluminum suspension.

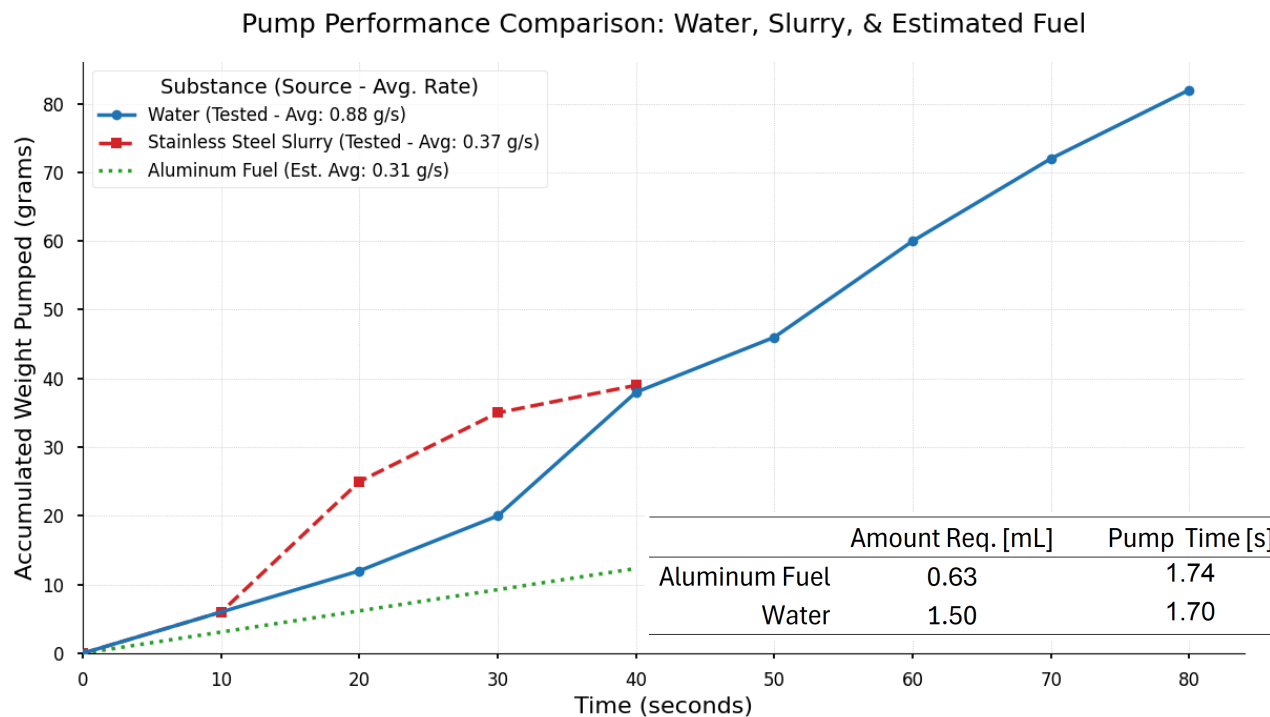


Figure 3.6: A table displaying the pump rates for each fluid with the accumulated weight pumped in grams shown over time in seconds. The table displays the calculated time required to pump the reactants based on the required reactant amount.

The results shown in Figure 3.6 show that both reactants will be pumped in approximately 1.7 seconds. The equal timing for both reactants means that the reactants should each run for the same amount of time allowing for laminar flow mixing at the y-joint and reaction initiation before dispensing into the bladder. This is the optimal scenario for reactant dispensing and should allow for optimal reaction kinematics and ideal gas generation.

3.3 Gas Bladders (Johan Thesis)

The subsystem which consists of one inflatable bladder and one gas pump, is our primary buoyancy alteration module, as the change in bladder volume directly alters forces on the engine test setup. There is also a check valve that allows for gas to escape from the flooded section. Throughout the testing cycle, this bladder also serves as storage for non-gaseous waste products of negligible volume to prevent mass loss.

3.3.1 Inflatable Bladder Design Overview (Johan Thesis)

One silicone rubber bladder made of a material very similar to PlatSil Gel-0020 (actual material is proprietary to the Rapid Liquid Print company which manufactured it), changes volume through the generation of compressible gasses such as in tubing directly connected to it [8,9]. Through this process, it effectively changes the setups' buoyancy. The bladder starts in an

initially compressed state induced by the gas venting pump. As the reaction occurs and the bladder is filled with hydrogen gas, steam/condensate, aluminum hydroxides, and mineral oil, the bladder is set to inflate without the need to stretch the rubber (passively) to a volume of 350 mL. As a safety measure, the bladder can passively hold 400 mL and can stretch to 964% times its size, being able to hold a maximum of almost 4 L [9]. This feature prevents failure through accidental excessive reactant pumping to the bladder while testing. This is based on the linear elongation rating of poured PlatSil and may not be the same for PlatSil geometries made with a 3D printing method.

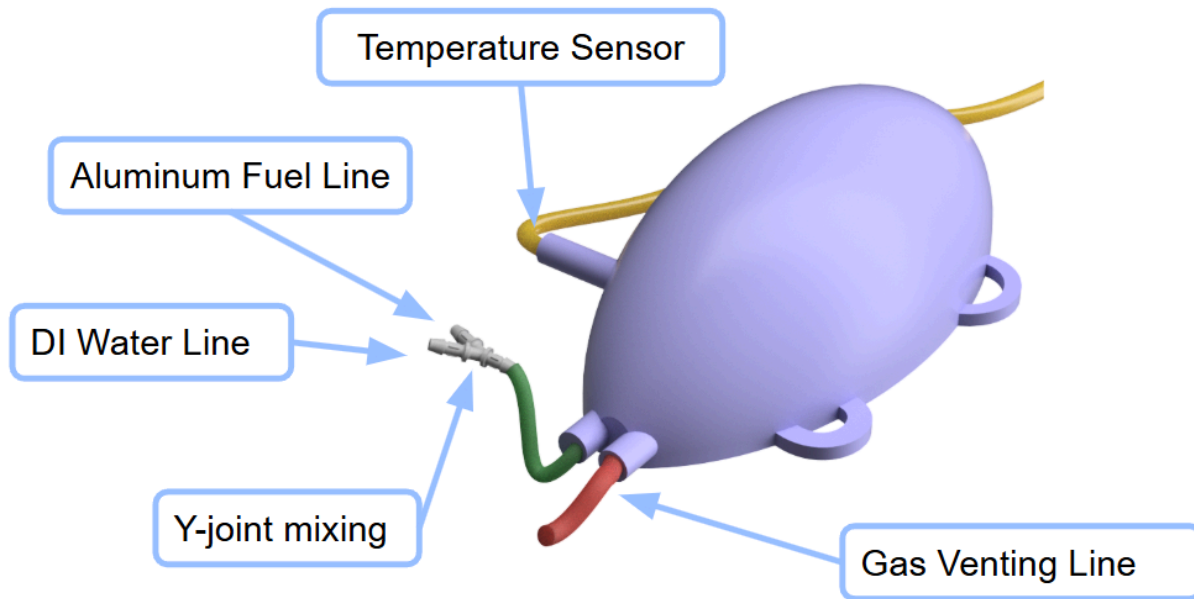


Figure 3.7: CAD of tested bladder module.

It was a critical requirement that the bladder's material must withstand a momentary peak reaction temperature of $\sim 200^{\circ}\text{C}$ and prevent outgassing. The bladder material can operate at a temperature range of -53°C to 232°C well within the desired operational range for this test setup and eventual full scale engine. The thermal properties of the silicone bladder were also analyzed via de creation of a simplified thermal model as shown in Figure 2.2 in section 2.2. It was important to minimize the heat transfer rate to the bladder. Because of this a shape similar to a sphere was used which minimized the surface area in contact with the interior while also meeting the flooded hull size constraints.

When the flooded hull is filled with incompressible water, the rubber's flexibility ensures that there is no pressure differential across the two sides of the bladder, even when steam condenses. Thus, we've assumed outgassing of hydrogen and steam to be negligible whenever the test setup is submerged in water. This bladder design does not consider potential outgassing if

the bladder is surrounded by air, hence we take the precaution to prevent H^2 leakage by filling the hull with water prior to starting any chemical reaction. 1/4 " ID Barbed fittings were chosen as the interphase between the system tubing and the bladder. These were chosen to reduce manufacturing complexity and make the design compatible with the RLP manufacturing process.

3.3.2 Inflatable Bladder Manufacturing Methods (Johan Thesis)

Manufacturing of the final test bladder, pictured in Figure 3.8, was carried on by the Rapid Liquid Printing (RLP) company [8]. This company specializes in 3D printing complex silicone shapes through the use of their "gravity free printing" technology, which injects silicone into a gel medium that holds it in place. RLP was chosen because of their willingness to fulfill low quantity orders at relatively low prices, as well as our desired customization capabilities. They quoted a \$350 initial charge for the production of one bladder and a \$100 charge for any subsequent order. This was a far cry from the thousands of dollars other companies quoted for an injection molded silicone bladder.



Figure 3.8: 3D printed silicone bladder

As good and affordable as the RLP bladder is, it has some issues. Mainly, the printing process struggles with making small hollow features such as the three 1/4" barbed fitting connections present in the bladder for tubing connections and for a temperature sensor. These connections had imperfections like the one shown in Figure 3.8 which led to weak points that failed with use. The bladder still remained airtight after rework but future work is recommended

to adjust the print orientation with RLP to avoid stress concentration around fitting connections. Additionally, it is recommended that the thickness of these connections is increased along with the barbed connection size and that a less hard silicone be utilized (10A shore hardness vs 20A shore hardness). In an attempt to address these issues (and given the uncertainty during RLP bladder manufacturing) other in-house manufacturing processes were attempted.

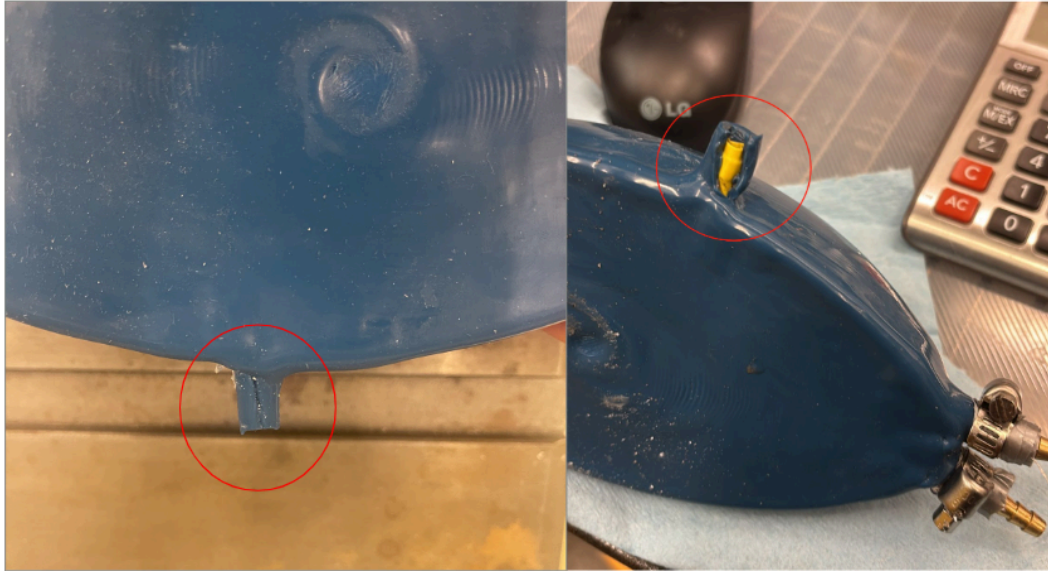


Figure 3.9: RLP Bladder fitting connection imperfection (left) and failure (right).

In an attempt to cast a silicone bladder in house, a three-part-mold was designed. It is pictured in Figure 3.9. It featured two outer molds made from PLA plastic and a removable inner mold. At first the inner mold was to be 3D printed out of PVA plastic, a water soluble plastic, with the idea that this plastic would be dissolved inside the bladder after the silicone cured and then removed. Unfortunately, PVA proved to be too brittle to be safely separated from the printing bed. After several print attempts, this material was switched for low temperature wax, which could be melted off of the bladder in an oven. A PLA mold was made for the wax inner mold and then cast. This proved too difficult to take off the PLA even with the application of mold release spray and thus was also abandoned.

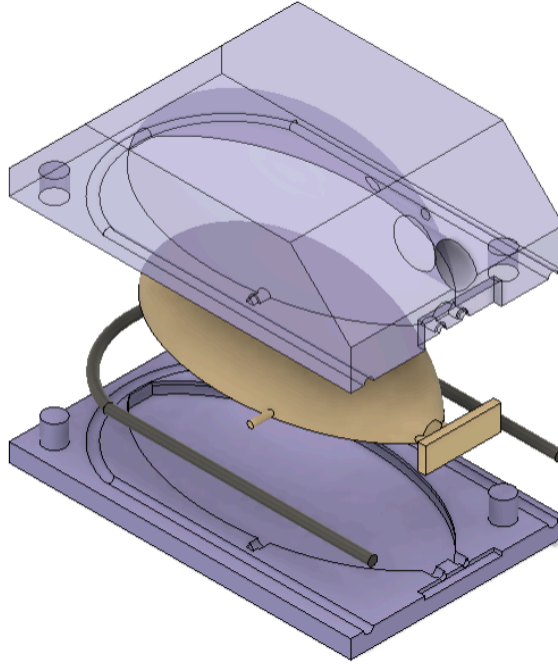


Figure 3.10: Three part mold for silicone casting. The two outer molds made from PLA are pictured in purple. The inner wax or PVA mold are pictured in beige. A piece PVC soft tubing is inserted into a groove in the molds to prevent leakage of silicone.

3.3.2 Check Valve (Johan Thesis)

One check valve, at the top end of the flooded hull, as pictured in Figure 3.11, was used to allow water and gases to exit the buoyancy engine when pressure builds up in the flooded hull due to bladder inflation. The use of check valves means that this process can occur while also prohibiting the flow of water back into the engine. The cracking pressure or pressure at which the valve opens must be high enough that it's not accidentally triggered unintentionally by small changes in flooded hull pressure but low enough that it can be triggered with a small reaction, thus improving fuel efficiency. Figure 3.12 shows this opening and closing process. Thus, a check valve with cracking pressure of 2 psi was chosen. This specific model, pictured in Figure 3.13, was also chosen for its compact size, high corrosion resistance of its stainless steel 303 body, and low cracking pressure of 2 psi [10]. It is paramount that the hull check valve has a minimum flow rate equal to 300 ml/min (Bladder Reaction Gas Flow Rate). This requirement was analyzed in the following calculation [11]:

$$Q = \frac{C_v}{\sqrt{SG/\Delta P}} = 0.1131 \text{ GPM} \sim 1014 \text{ ml/min} \quad (3.4)$$

For a cracking pressure of 2 psi (P), a check valve flow coefficient of 0.19 (Cv) and a working fluid with specific gravity of ~1.0 (SG, fresh water),[insert reference] this check valve will permit hull water to flow at a rate of ~1014 ml/min, well above our 300 ml/min requirement.

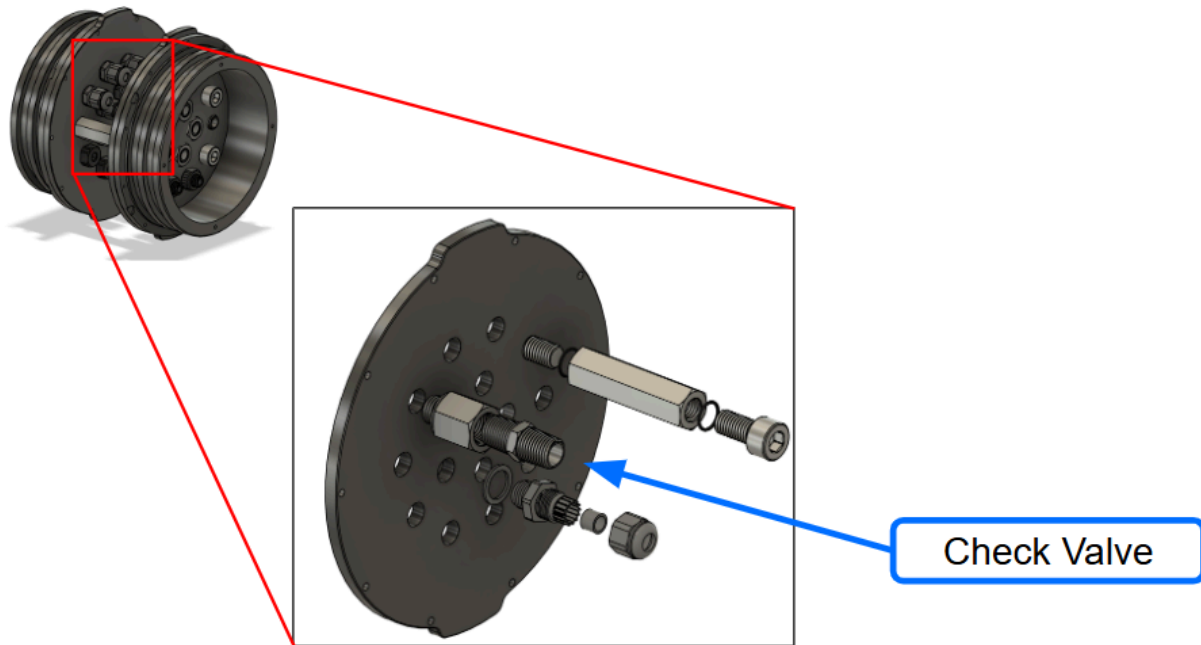


Figure 3.11: The placement of the check valve relative to the flooded hull section is shown above.

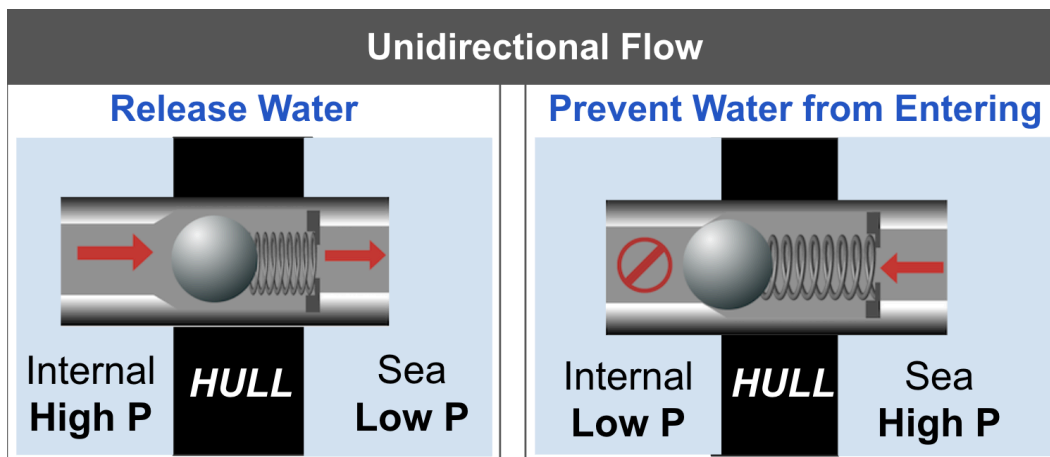


Figure 3.12: The opening and closing process of a check valve is shown above. It can be seen in case A that when the water pressure equals the cracking pressure in the correct opening direction (in this case to the right) the check valve will open and let the working fluid through. In case B, since the pressure difference is in the incorrect direction, no working fluid will flow through. This will be the case for any pressure below the maximum working pressure of the check valve, at which point the check valve will fail [12].

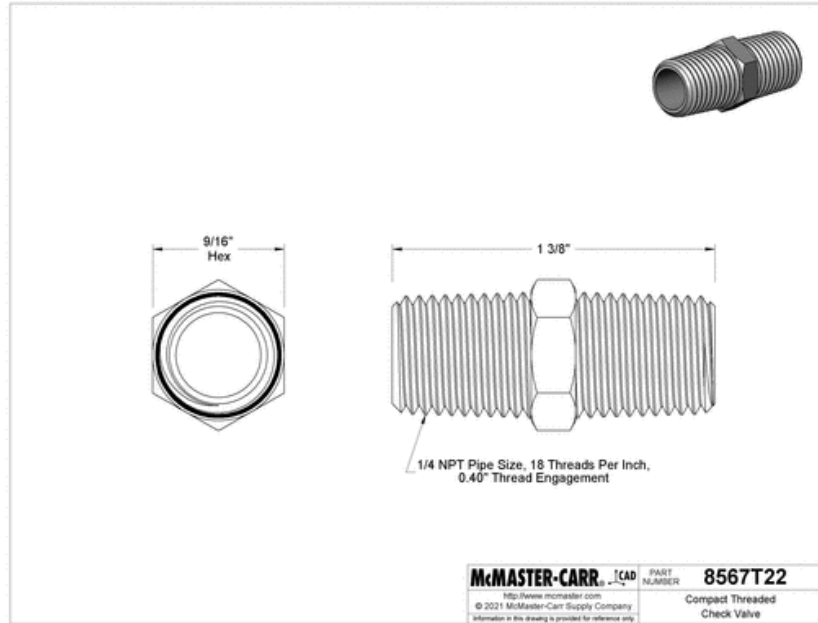


Figure 3.13: Front and side views of the “compact Threaded Check Valve with Loctite® Sealant, 303 Stainless Steel Body 1/4 NPT Male” sourced from McMaster is shown.

During testing it was observed that not all of the vented gas from the bladder would exit the flooded hull while the engine floated parallel to the surface. It was concluded that this problem was caused by improper placement of the check valve. For future iterations of the test setup multiple check valves should be placed radially throughout the midsection of the flooded hull enclosure. This will ensure proper venting through the valves regardless of pitch and spatial orientation.

3.3.3 Gas Pump for venting (Johan Thesis)

A Gas pump held in the electronics section is connected to the gas bladder and serves to vent the gasses into the flooded hull section of CALIPSO. The Gas venting pump primarily serves to move water out of the bladders when transitioning from ascent to descent in the dive cycle. Their secondary purpose is to vent gas during ascent to counteract acceleration to the surface from bladder expansion as water from the flooded hull exits the check valve. Both pathways are described below.

When CALIPSO is ready to descend—at the surface or at any point during ascent—the gas venting pumps remove nearly all the gas from the bladders. To create the necessary pressure differential across the hull to crack the check valve, the sea water pump is then used to pump in water which increases internal hull pressure. The gas is then released when the check valve opens, and the gas-to-water volume ratio decreases, achieving a negative buoyancy and thus descent. This gas venting pathway is illustrated in Figure 3.14.

Key functional requirements used to select the gas pump for bladder venting include unidirectional flow, seal, size and maximum pressure head. The unidirectional flow is important to ensure that only gas is flowing out of the hull, and water is not flowing in. The seal is important because gases are prone to leakage, especially the small hydrogen gas molecules. Size was the greatest constraint, as the pump needed to fit inside the electronics enclosure. Piston pumps and diaphragm pumps are two common types of gas pumps. Diaphragm pumps tend to be a little smaller and are known to hold a very good seal. The diaphragm material also acts as a boundary layer between the passing fluid and internal mechanism of the pump, so diaphragm pumps are more compatible with corrosive materials [13]. This is important for longevity as hydrogen gas is corrosive to metals [14]. A relatively high maximum pressure head was important to ensure that the gas bladder would collapse completely.

Functional requirements that are notably not listed here include the ability to be submerged in fluid, temperature rating and flow rate. The pumps' temperature rating was also of low concern since the fluid passing through the pump, the hydrogen gas, will have already cooled within the bladder by the time venting would take place. Flow rate similarly isn't critical as the testing device is moving slowly and changing buoyancy is not time-length-sensitive. Additionally, these pumps do not need to work against a great pressure differential as the bladder and flooded hull are in pressure equilibrium. The negligible pressure differential between the flooded hull and the bladder also minimizes concern for hydrogen leakage.

CALIPSO uses WP-370B Diaphragm Pump for Gas to vent the gas bladder pictured in Figure 3.14. This pump was selected as it met our requirements and was very affordable [15].

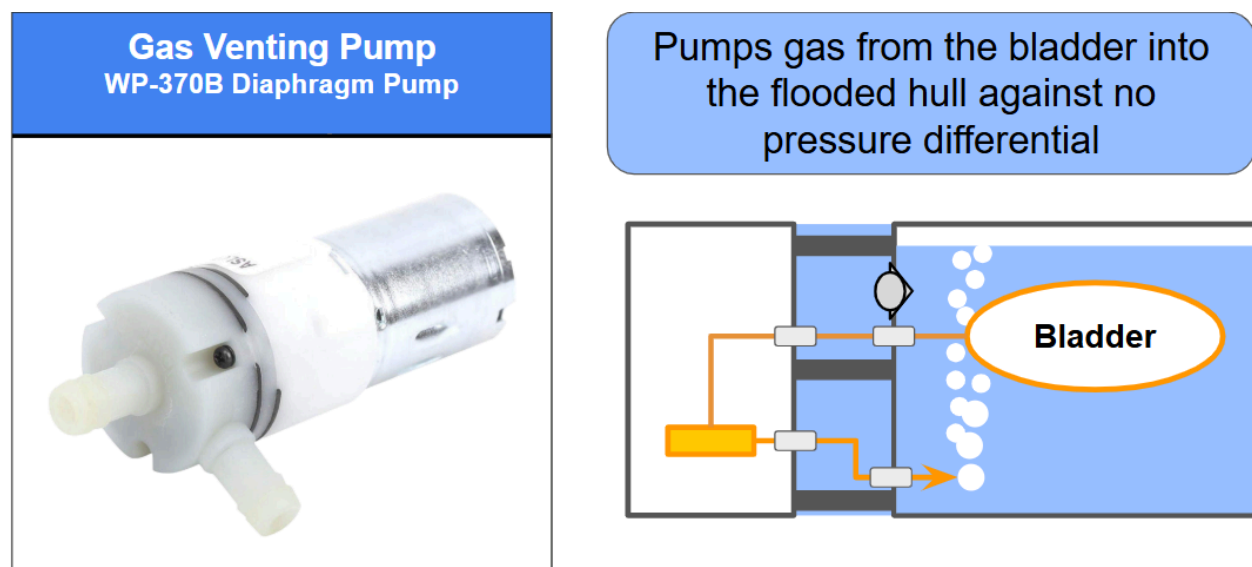


Figure 3.14: The selected gas venting pump is shown alongside its working diagram.

Flow rate testing was performed to ensure that the pump could collapse the bladder in a reasonable time. After several trials the pump managed to vent an average of 9.4 mL/s of air.

This translates to a bladder venting time of ~ 42 seconds which is a reasonable time to wait in between cycles.

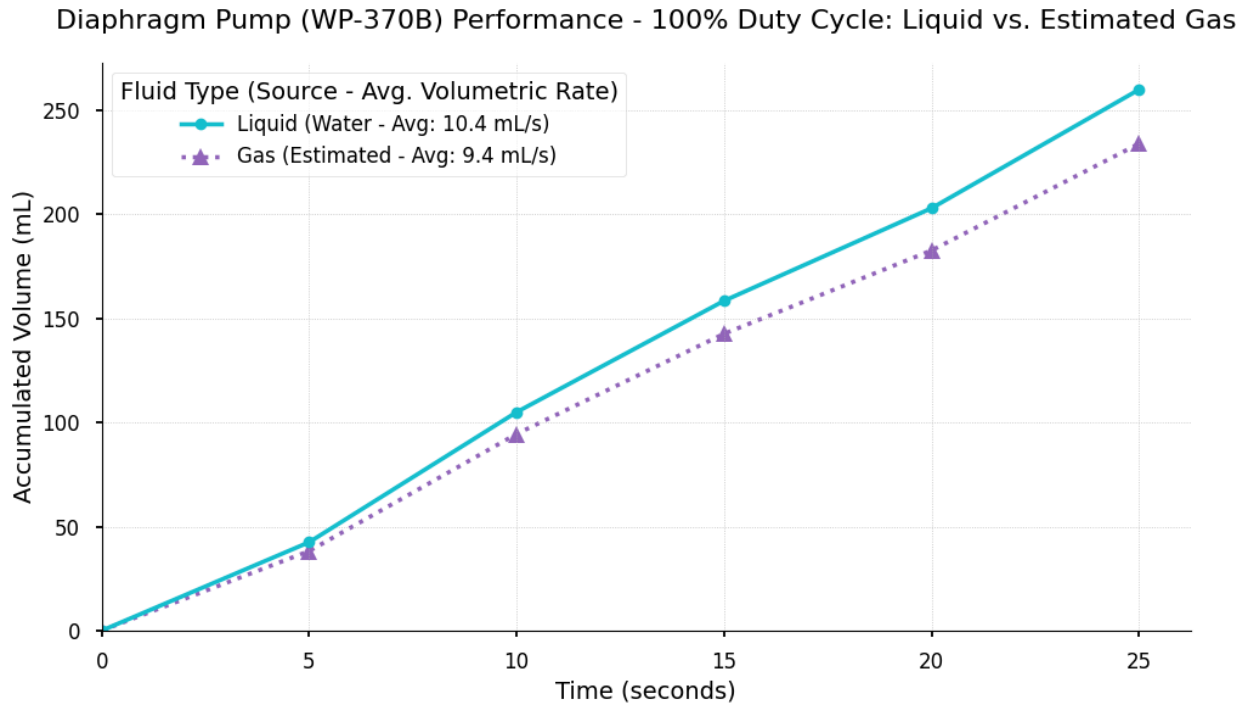


Figure 3.15: Diaphragm pump performance testing results for water and air working fluids.

3.4 Enclosure and Mounting System

The system we initially started working on in February consisted of a single enclosed tube containing a flooded hull section and an electronics enclosure filled with mineral oil to enhance sealing at a 500 m dive, as indicated in Figure 1.2. However, due to the need of descoping, the team's approach to overall housing evolved significantly. Critical changes were primarily driven by the reduction of operational depth to 5 meters and the removal of fine buoyancy control and moment balance control features. Design modifications regarding the enclosure included reducing the number of pumps and consolidating the system into a single reaction bag instead of two. The previous design required a waterproof interface not only with the outside environment but also within the tube itself, between the electronics enclosure and the flooded hull.

The internal waterproofing posed significant concerns during manufacturing and installation. To mitigate leakage risks, the system was restructured into two independent tubular sections, as illustrated in Figure 1.1 and 3.16, both of which were sourced from a commercial-off-the-shelf enclosure from Blue Robotics. Each enclosure consists of an acrylic tube, flanges with O-rings, and acrylic or metal end caps that screw onto the flanges. The right

tube in the figure below shows the electronics housing, while the left contains the flooded hull that houses the reaction bladder, fuel, and ammonia bag. These enclosures are rigidly connected via steel coupling nuts. Rigid tubes passing through cable glands were sealed with teflon tape and grease to facilitate the flow of various fluids, such as water, ammonia, and gas, between the two sections.

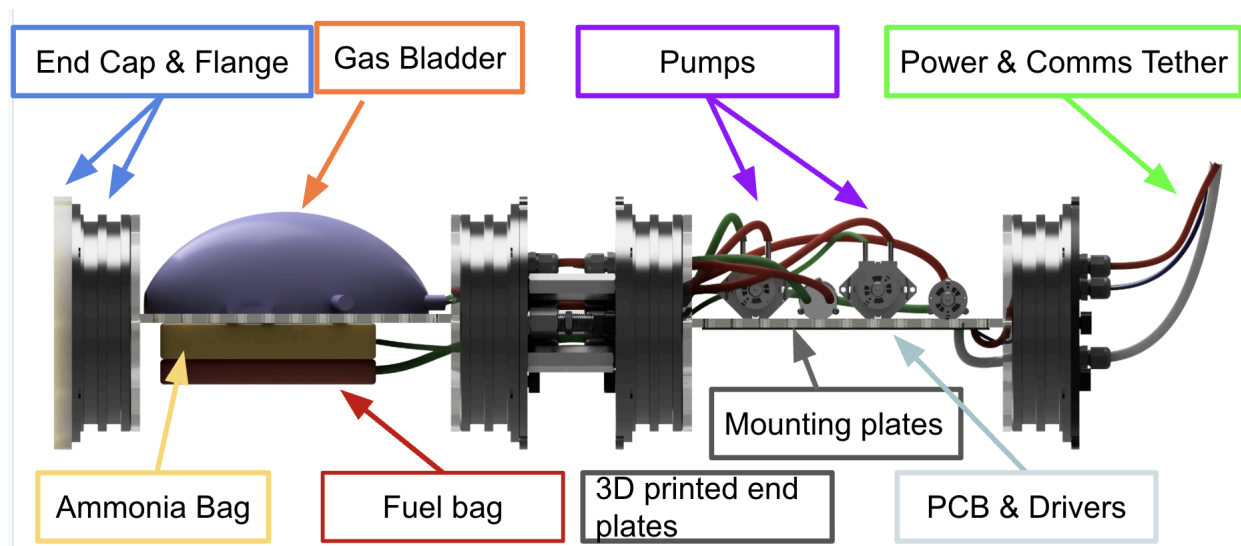


Figure 3.16: enclosures for electronics and bladder with components

3.5 Assembly

3.5.1 Reaction Pipework (Johan Thesis)

Reaction pipework routes reactants/products throughout our engine and is the location where the chemical reaction takes place. To inflate the gas bladders and cause a change from negative to positive buoyancy, the proper volumes of aluminum liquid fuel and D.I. water are pumped from storage bags in the flooded hull compartment into a steel Y fitting in which the reactants mix, producing hydrogen gas and steam from the reaction heat. The evolution of high pressure gases discharges all fluids, any unreacted aluminum particles, and aluminum hydroxide waste particles from the tube into the bladder, where the reaction continues. Figure 3.17 shows a schematic of the pipework from the reactant bags, through the pumps, and into the bladder, and how the tubing will be connected with selected fitting types based on selected tubes and dimensions. The reactant merge point is located outside of the electronics enclosure in order to protect the electronics from the heat of the reaction.

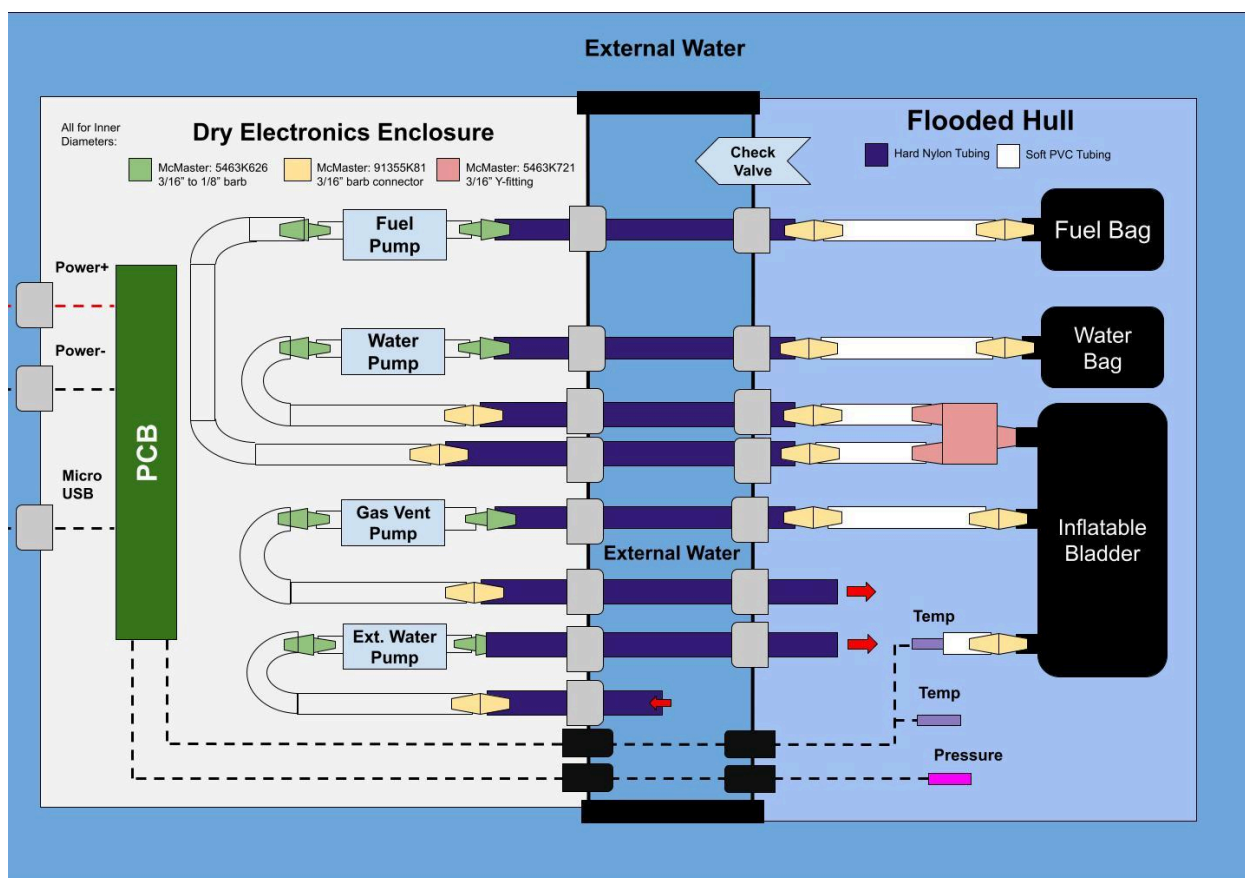


Figure 3.17: Reaction piping schematic with tubing dimensions and fitting types to use.

High-temperature Viton® fluoroelastomer opaque soft tubing (3/16" ID, 5/16" OD) was chosen for the reaction tubes in which the reaction occurs (tubing from the point of reactant combination onward) for its extreme corrosion resistance, temperature resistance (-10 F - 400 F) and largely reduced weight compared to metallic options [16]. General non-reaction tubing in the test setup, namely for flows through all pumps in the electronics enclosure, was chosen to be soft PVC tubing (3/16" ID, 1/4" OD) [17]. This tubing was chosen for its affordability, excellent chemical and corrosion resistance and high bendability. Since there are many inlet and outlet pump lines in the engine that will need this tubing, high flexibility is a huge advantage to allow tubing to be routed within the cramped spaces of the engine, especially its electronics enclosure. All tubing that goes across the cable glands was chosen to be hard nylon plastic tubing (3/16" ID, 1/4" OD) for its compatibility with the cable glands [18]. This is because cable glands require hard tubing in order to properly seal and guard against water leakage at pressure. All barbed fittings were sourced to be either 3/16" ID brass barbed fittings or 1/8" to 3/16" ID brass barbed fittings for their affordability and compatibility with the integrated pump tubing [19,20].

3.5.2 Electronics Enclosure and mounting

The electronics section features a horizontal mid-plate that is stabilized in the center of the tube by joints connecting to 3D-printed end plates resting inside the flanges. This mid-plate is fastened to the 3D-printed caps using box joints. It plays a crucial role in securing all the pumps within the pcp. Additionally, the holes on the mid-plate facilitate the attachment of zip ties to keep all components securely in place.

The laser-cut acrylic midplate with perforated holes and the 3D-printed end caps underwent several versions during testing. The 3D-printed caps fit inside the flanges on the inner side and are secured with dowel pins. We modified the design multiple times to facilitate assembly and disassembly. The 3D-printed end caps were initially designed as shown in Figure 3.18, with perforated holes that tightly fitted the acrylic perforated plate on both sides. During testing, the snug fit worked on one side of the enclosure; however, while closing the corresponding other side, the resistance of the O-rings attached to the flanges made it difficult to align the holes in the 3D caps with the acrylic mid mounting plate. As a solution, we changed the joint sizes for the mid plate. We completed three iterations. In the first iteration, the joints were narrower, resulting in a tight fit on one side and a looser fit on the other. For the third iteration, we extended the teeth of the mounting plate because we continued to face challenges in aligning the holes of the 3D-printed plate with the mounting plate.

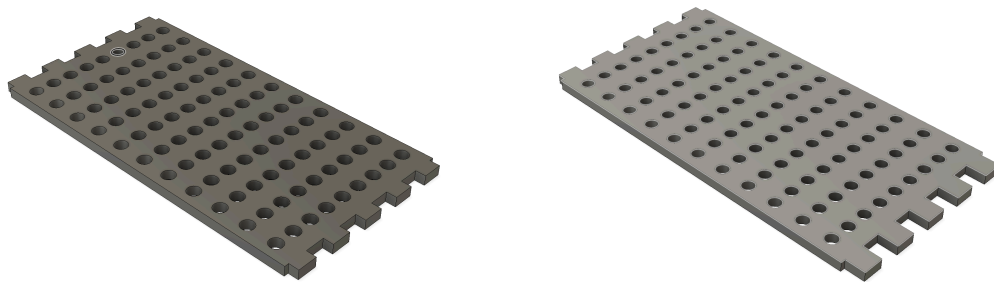


Figure 3.18: Mid plate versions. The initial version is on the left, and the revised version is on the right with longer joints.

Additionally, the design of the 3D-printed end plates complicated the tubing process, as the tubes needed to pass through each hole. So the second iteration of the end caps features open half-circle transition holes, allowing more flexibility for the tubes to pass through.

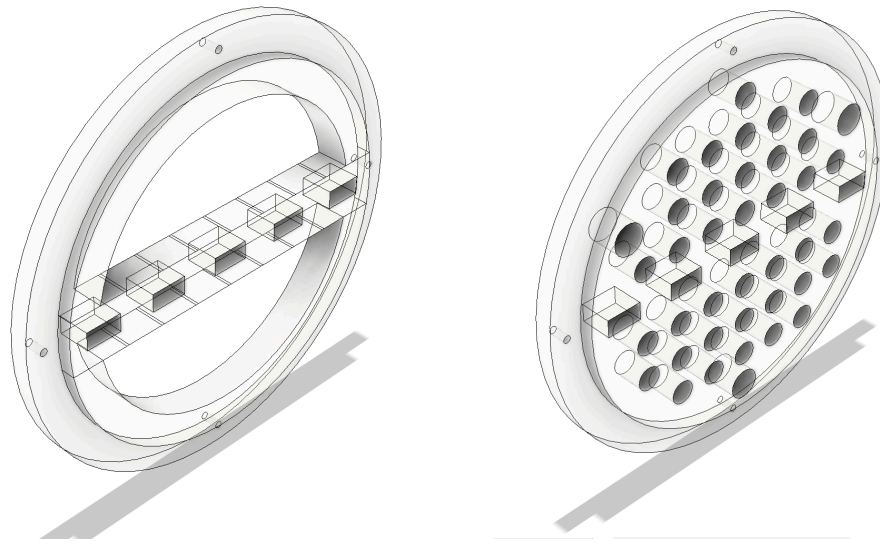


Figure 3.19: 3D printed end cap with perforated holes on the left and cut outs on the right.

3.5.3 Mounting for Hull section & Refueling

The hull section was initially designed similarly to the electronics enclosure, featuring an acrylic mounting akin to the electronics plate that secures all system components. This plate is positioned horizontally between two vertical 3D printed end plates, which are held in place using box joints. The end plates sit within end caps, attached to the flanges on the inside with dowel pins. This design encountered several challenges discussed in the electronics section in 3.6.2. Similar revisions were applied to this hull section; however, since it accommodates the bladder, fuel, and ammonia bags, aligning the 3D printed plate with the mid plate and securing it without moving the bags proved more challenging. Additionally, the assembly and disassembly of this design made it difficult to remove and attach all the bags during testing and refueling. To simplify refueling, we introduced a third version where the middle plate is attached to the 3D printed caps, allowing everything to be printed together. This approach facilitates the assembly process, enabling the acrylic tube to fit the 3D printed bracket effortlessly.

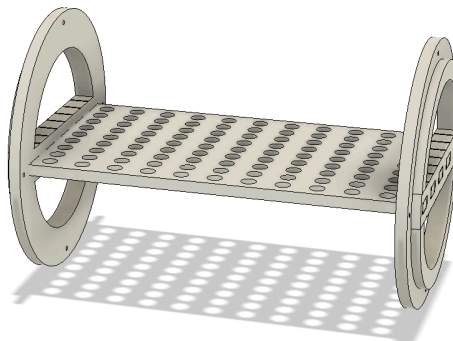


Figure 3.20: 3D printed bracket that holds the bladder, fuel bag and ammonia bag.

The system was designed to enable rapid prototyping and easy refueling. To refuel, operators simply slide out one end of the end cap and acrylic tube to remove the water and aluminum fuel bag, as depicted in Figure 3.21. It's essential to clear all reaction byproducts and flush the fuel path with oil each time to ensure safety. This design significantly reduces refueling and troubleshooting time by eliminating the need to disassemble the central connection—an area critical to waterproofing and prone to leakage with repeated handling.

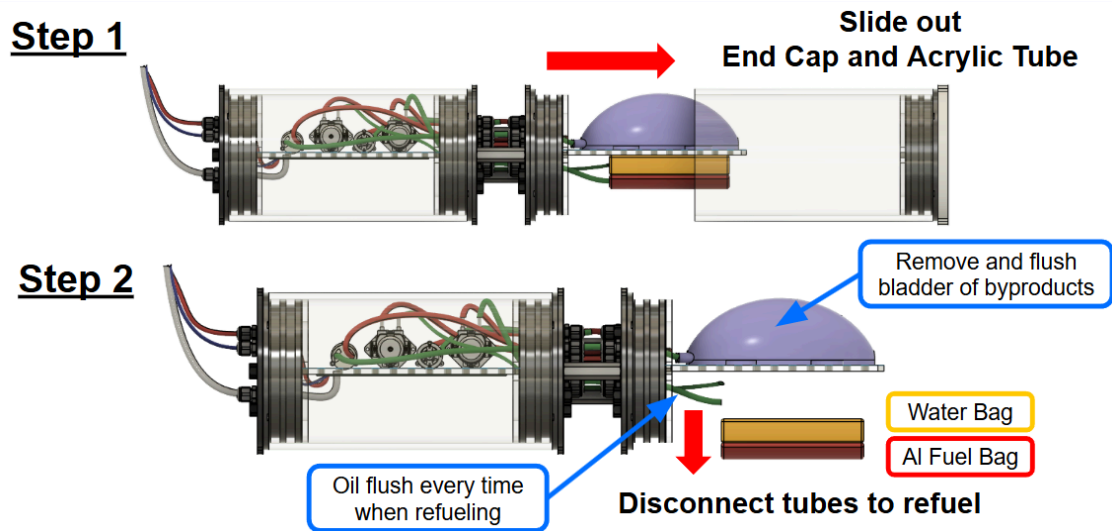


Figure 3.21: Detailed procedure schematic for refueling process

3.5.4 Connections Between Hull and Electronics Sections.

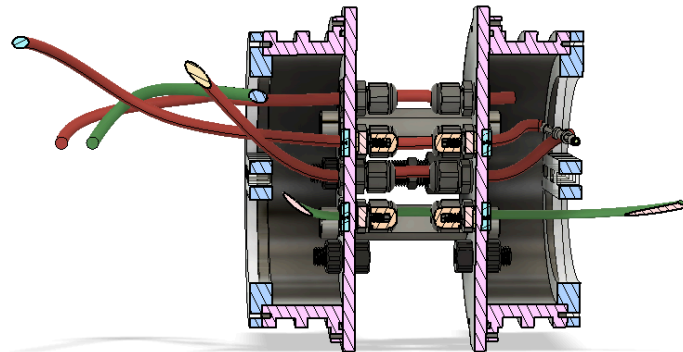


Figure 3.22: Section view of connections between the Hull and the electronics. The rigid connectors and the cable gland are visible.

The two enclosures are securely connected using steel coupling nuts. Rigid tubes, which go through cable glands, are sealed with epoxy to enable the flow of various fluids like aluminum paste, water, and gas between the two sections. During the first iteration, PG7 plastic cable glands were utilized alongside soft, flexible tubing. However, this design encountered waterproofing problems as bending of the tubes caused water to seep in and out of the threads of the plastic cable glands. To address this, we switched to metal cable glands and rigid tubes for better connectivity between the two sections. Moreover, all the threads and O-rings of the glands were lubricated, and all threaded parts were wrapped in Teflon tape.

3.5.5 End Cap Design and Further Waterproofing Details

We pre-ordered metal and acrylic end caps from Blue Robotics, which connect to the flanges using M3 screws. However, due to the number of required connections and the size of our cable glands, we determined that manufacturing the end caps in-house was necessary. We recut all end caps designed for cable or tube connections using a water jet from 3.175 mm aluminum. By replicating the Blue Robotics profile shape of the cap, we incorporated additional holes and better arrangement. The caps were attached to the flanges with the same screws, but since we used thinner aluminum, we had to add five washers to each screw before securing them to the flanges.

During testing, we had issues closing the acrylic tube firmly due to pressure build-up on the hull side, so we added tabs on top of the end caps. These tabs had 10 mm holes that threaded rods went through and secured the end caps closed using nuts on each side. We were able to add the tabs that held the rods on the metal end caps because we had the extra space from the washers we had added in the previous version. However, we were still using the acrylic cap on one side of the hull from Blue Robotics, which didn't have an additional gap with washers. So, we had to recut this acrylic cap from thinner acrylic.



Figure 3.21: Tabs and threaded rods mounted on both sides of the flooded chamber to resist pressure pushing out the end cap

3.6 Electronics & Controls

The CALIPSO electronics and controls subsystem achieves precise reactant dispensing of aluminum paste and desalinated water reactants, water intake into the flooded hull, and gas venting of the internal bladder. CALIPSO uses a combination of four unidirectional pumps and one check valve to route gases, reactants, and seawater into and out of the flooded hull and internal bladder. The pumps are powered by a 12V tether connected to an external power supply, which, with the use of buck converters, would be replaced by the typical 24V power source onboard a UUV. The pump motors are actuated by an ESP32 microcontroller whose external electronic components are housed on a specially manufactured PCB inside the CALIPSO electronics housing.

3.6.1 Controls

3.6.1.1 Fuel Dispensing

The reactant management system aims to reliably store reactants within the flooded hull of the enclosure, deliver and mix precise amounts of reactants ensuring predictable reaction rates, and utilize the gas products from the reaction to displace water volume within the hull and thereby generate positive buoyancy.

Once flexible storage bags were sourced and integrated for reactant storage, the task of delivering and mixing precise amounts of reactants was solved through a combined mechanical and controls team effort. A peristaltic pump, namely the Gikfun dosing pump, was selected to pump the aluminum paste, due to its high average fuel particle size. Even mixed with a higher viscosity oil, the peristaltic pump was necessary to convey the coarse mixture. The two reactants mix at a y-joint and the reaction begins, continuing into the downstream tubing. The Gikfun dosing pump was also selected to pump the DI water reactant. The peristaltic pumps are ideal for the reactants' small dosing amounts, for lack of contamination, and for chemical compatibility.

CALIPSO's designed bladder readily expands as the reaction produces hydrogen gas. This will increase the pressure in the chamber until it reaches the cracking pressure of the check valve that regulates flow between the hull and outside environment. Once this pressure is reached the check valve opens and the water is pushed out of the hull until the pressure within the hull equals the outside pressure.

A manual open loop control system was used for the actuation of the two peristaltic pumps described above. Pulse width modulation (PWM) was used to allow for varying the rate of material dispensing based on pump needs. Through pump testing with the reactants and stand-in mixtures, the ideal duty cycles were determined to be 35% for the water reactant pump and 35% for the aluminum paste pump.

Using the ASCEND command on the graphic user interface seen in Figure 3.22, the CALIPSO user chooses to dispense the exact amount of reactants needed to cause a change in buoyancy. The calculated aluminum paste to water reactant ratio is 0.42:1, and the corresponding reactant amounts are 0.21 mL and 0.50 mL, respectively. The determined reactant amounts correspond to a dispensing duration of 0.57 seconds for the water reactant pump at 35% duty cycle, and 0.58 seconds for the aluminum paste pump at 35% duty cycle. The aluminum and water reactant dispensing will both end at effectively the same time, allowing for laminar mixing in the y connection of their tubing and in the bladder. At any point, the user can block either or both reactant pumps, effectively being able to inject any desired amount of reactant to create any desired amount of hydrogen gas. This approach is straightforward but still ensures precise reactant delivery.

3.6.1.2 Bladder Venting

In order to support the descent operation, the bladder needs to be fully vented of hydrogen gas before seawater can be introduced to fill the flooded hull. A diaphragm pump, namely the WP-370B diaphragm pump, was chosen to vent the internal bladder of all gases, including hydrogen gas and steam. Diaphragm pumps are ideal for the higher flow rate of bladder venting, for higher pressures, and for self-priming.

A manual open loop control system was used for the actuation of the bladder gas venting pump. Again, PWM variation allows for differing rates of gas venting. The amount of gas necessary to expel, in order to almost completely vent the internal bladder, was calculated to be 543 mL. This amount of gas corresponds to a dispensing duration of 58 seconds at 100% duty cycle. At any point, the user can block the gas venting pump, effectively being able to keep the bladder at any desired level of gas volume, which can allow for intermediate or gradual buoyancy shifts.

3.6.1.3 Hull Flooding

In order to support the descent operation, a diaphragm pump, the WP-370B diaphragm pump, was chosen to intake seawater into the flooded hull. Similarly to bladder venting, a diaphragm pump is ideal for the hull flooding higher flow rate and pressure. Once the internal bladder is vented of gas, sea water is pumped into the hull and CALIPSO experiences a net negative buoyancy change, causing the engine to descend. This water intake pump completes its task most efficiently when injecting the seawater at the highest flow rate supported.

As the others, a manual open loop control system was used for the actuation of the water intake pump. Again, PWM variation allows for differing rates of water intake based on desired buoyancy changes. However, the ideal duty cycle remained at 100% to support maximum flow rate. The volume of seawater necessary to satisfactorily fill the flooded hull and achieve a buoyancy change was calculated to be 543 mL. This full flooding volume corresponds to a dispensing duration of 52 seconds at the chosen 100% duty cycle. At any point, the user can block the water intake pump, effectively being able to inject any desired amount of seawater into the flooded hull to achieve any desired negative buoyancy change.

3.6.1.4 Logic and Sensing

The system relies on time-based sequencing and predefined actuation commands rather than sensor-driven feedback. External commands provide triggers for each phase of the cycle. These commands consist of ASCEND, DESCEND, ABORT ASCENT, and ABORT DESCENT. Code for this logic was developed within the Arduino IDE, chosen for its user-friendly interface and easy deployment. This code sequences the water and aluminum paste dispensing, gas venting, and water intake, with an ESP32 microcontroller delivering the digital outputs. Abort mechanisms are included in the event of hardware malfunction, unexpected sensor readouts, and for general convenience of pausing without severing communications. Using serial communication, two graphical user interfaces (GUIs) were developed for convenient control and monitoring during testing. The first GUI is the Commands Dashboard, as shown in Figure 3.22, which connects to CALIPSO's internal ESP32 and holds operation triggers and status readouts.

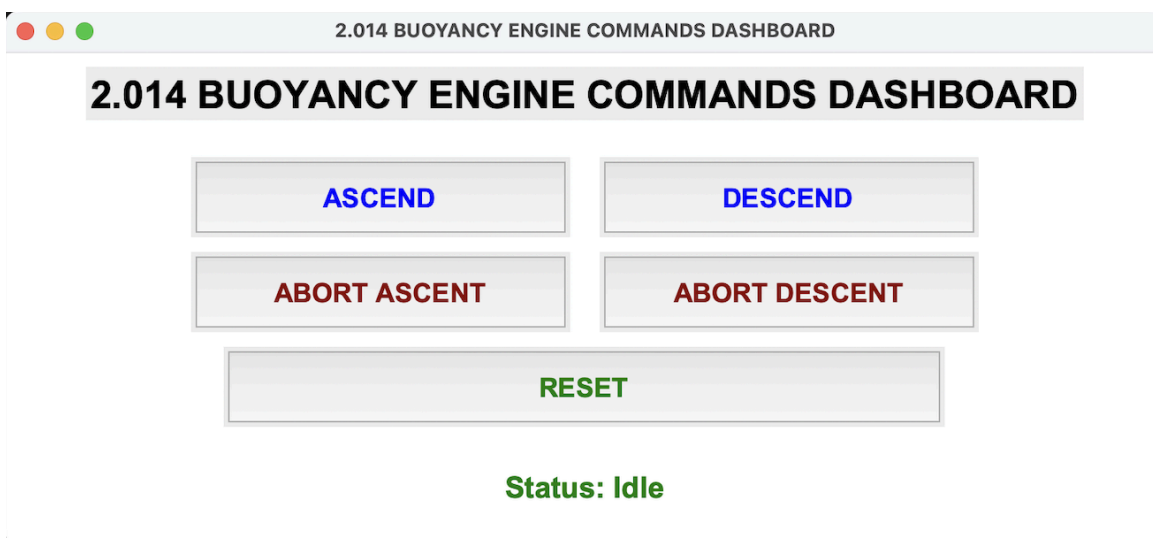


Figure 3.22: CALIPSO Commands Dashboard.

During operation, hull pressure, internal bladder temperature, and external temperature can be monitored in real time to confirm when a reaction is happening. The pressure and temperature sensors serve as passive monitoring tools to verify expected system behavior, though they do not actively influence control decisions. A Blue Robotics pressure sensor is placed in the internal hull to measure pressure increases due to the chamber filling or the bladder inflating. One thermistor placed inside the bladder checks for temperature increases due to the reaction, while another thermistor inside the hull acts as a temperature setpoint to reduce noise in sensor readings. The second GUI is the Sensors Dashboard, as shown in Figure 3.23, which connects to CALIPSO's external Arduino Uno and shows all sensor readings. The Sensors Dashboard includes real time graphs of internal bladder temperature ($^{\circ}\text{C}$) versus time, as well as hull pressure (hPa) versus time. All sensor readout data is automatically saved into a csv file once the session is complete, and both the commands and sensors dashboard can be opened on one device at the same time. All CALIPSO scripts can be found in the team Github [21].

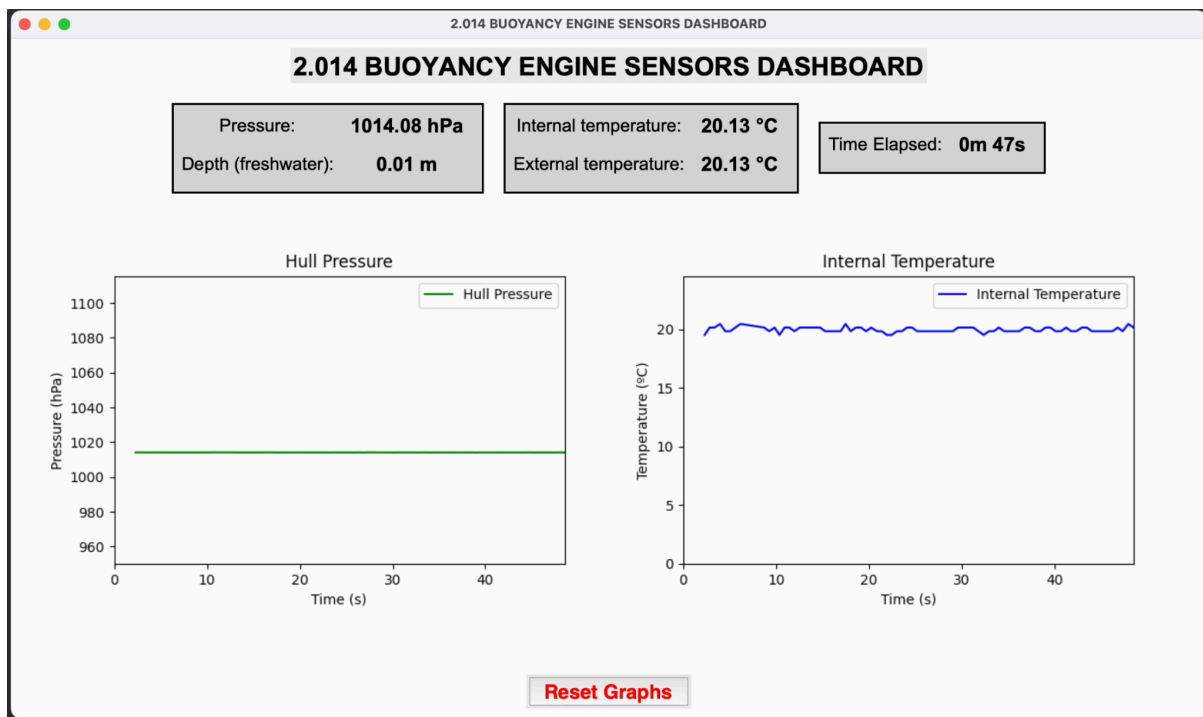


Figure 3.23: CALIPSO Sensors Dashboard.

3.6.2 Electronics

The electronics served as the major component to integrating our control algorithm into the system. Due to the need for rapid prototyping and long lead time, we were not able to get the PCB on time so a breadboard setup was constructed to emulate the full functionality of the PCB design. The PCB CAD is depicted in Figure 3.24.

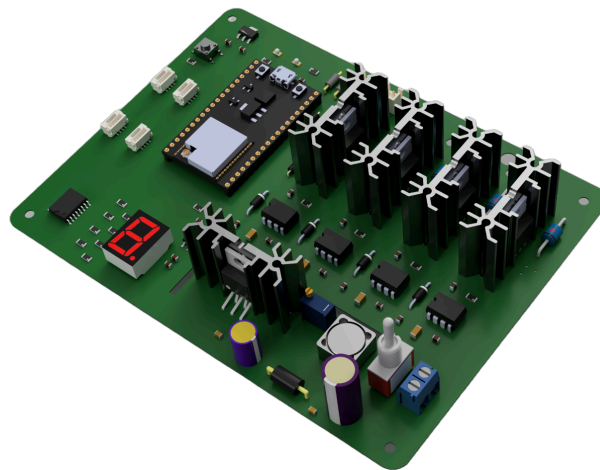


Figure 3.24: CALIPSO PCB Design

3.6.2.1 Operational Requirements

Due to our descopeing for validation of fundamental operational principles, the four motors require only unidirectional speed control. All motors run at 12V, with maximum continuous amperage expected to be around 0.5A for the DI water pump and the two diaphragm pumps, and 1A for the Gikfun dosing pump that handles the aluminum and oil mixture paste, due to the higher viscosity. This adds up to 2.5A or 30W of total power use during full operation. A thermistor and pressure sensor were chosen for the system. These serve to monitor the reaction, signaling the state of both inside and outside the reaction chamber. The temperature range necessary to be measured with the thermistor was from 20 °C up to 150 °C maximum. The pressure sensor uses I2C communication. Power levels of 12V, 5V, and 3.3V are required, with the 12V as the motor power, 5V as sensor or serial communication power, and 3.3V to power the microcontroller when USB is untethered. Furthermore, the input power must be able to traverse a multiple meter long cable without excessive power loss.

3.6.2.2 Motor Control

For controlling both the motors and sensor, an ESP32 was chosen, both for its small footprint and versatility for testing purposes. This also has the minimum number of output channels required for control of all peripherals on the board.

For the motor control, an IR540N MOSFET paired with a IR2125 gate driver was chosen. The IR540N is capable of significantly higher power requirements of up to 20A / 50W continuous, as well as operation up to 100V, so there was no concern in requirements here. With this in mind, this component was still chosen for comfortable thermal limits due to the low on resistance of 0.044 ohms, as well as long term reliability and ease of access for testing purposes. However, the gate to source voltage required for the IR540N to reach full saturation is 10V, which means that the 3.3V logic level from the ESP32 cannot be connected directly to the MOSFET. For this purpose, a gate driver is used to drive a bootstrap circuit and achieve a sufficient VGS. This works by charging a capacitor to 11.3V through a diode while the MOSFET is off, then using stored charge to power the gate. This limits the on time possible after each rising edge on the control signal, since the capacitor charge drops immediately, making PWM necessary at all times. At the same time, charging the capacitor takes time, requiring a minimum amount of off time. A 1 microfarad capacitor was chosen for this purpose, as it provides an on time of 4 milliseconds and a minimum off time of 50 microseconds, which allows for PWM signals down to 250 Hz with a maximum duty cycle of 98.75%. With the control signal for motor operation kept higher at 1k Hz, maximum duty cycle is 95%.

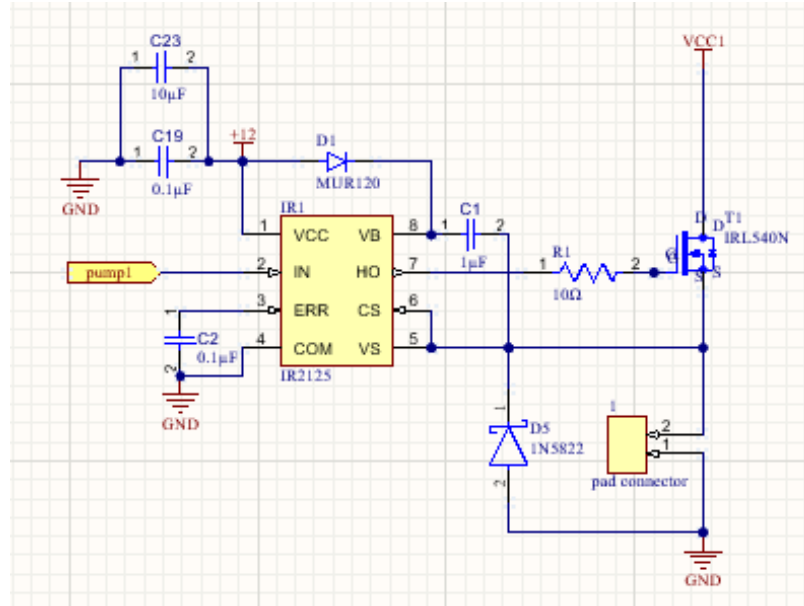


Figure 3.25: Pump Circuit Design

3.6.2.3 Sensing

An NTC thermistor was chosen to monitor temperature, which has a reference reading of 100 K at 25C. This provides the benefit of single port readings with an inbuilt ADC, and easily adjustable sensitivity by swapping the pull-down resistor value. The circuit was configured to have the thermistor on the high side of a voltage divider, which means as resistance values fall with temperature increase, the signal voltage increases in kind. To maximize the ADC resolution while maintaining the required temperature range, the pull down resistor was chosen so that the signal would reach 3.3V at exactly 150C. Using the Beta Model Equation, assuming a beta value of 3950K, we can predict the resistance value of the thermistor at 150 degrees C to be around 2k ohm. With a pull down resistor at 4k ohm, this would produce a signal of $5 \cdot (4/6) = 3.33V$, making 4k ohm the ideal value.

The Blue Robotics Pressure Sensor operates using an I2C interface. The necessary connector for this was included in the PCB, and routed directly to the ESP32 as external peripherals are not necessary.

3.6.2.4 Power/Thermal

In order for the buoyancy engine to be able to handle a range of input voltages, the LM2576 buck converter was included. This provides a steady 12V power source with inputs ranging from 15V to 40V. For a 5V and 3.3V logic source, the 7805 and 1117 linear regulators were used respectively. In addition, jumpers are included to allow direct 12V power input. During testing, the power was provided directly by a tether carrying 12V, and logic power was provided by a usb cable. The tether, while the full length was ultimately not necessary for testing purposes, 18AWG wire was selected to ensure minimal voltage drop across a 10m span. The IR540s running 2.5A output only 0.275W, each raising up to at most only 30-40C. However, as the

electronics must operate inside an enclosed sealed chamber, heatsinks were made to be significantly larger than would be necessary in air.

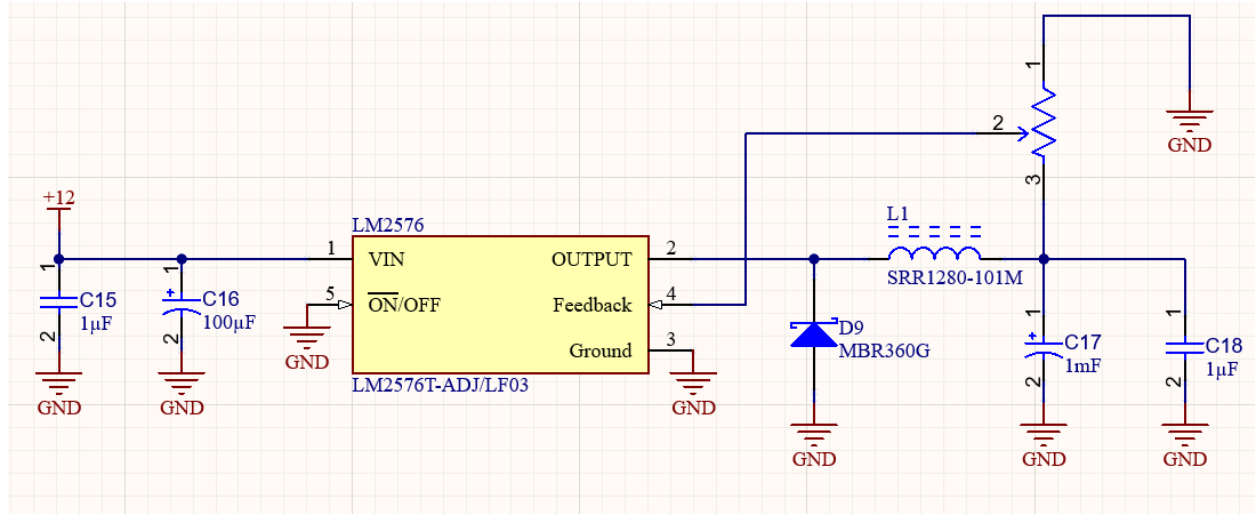


Figure 3.26: Power supply circuit design

3.7 Test Setup

A separate physical assembly was needed to hold CALIPSO and obtain measurements on its performance. This served two important functions, both the ability to collect results, but also as a means to quickly test system waterproofing.

3.7.1 Testing Needs and Concept

Several factors were useful to characterize the performance of CALIPSO, namely: total change in buoyancy, temperature of reaction, timing of reaction, and pressure over time. All of these metrics help in improving the understanding of the reaction and how they change when moved to a buoyancy engine compatible environment. A testing concept was developed from which these metrics could be measured, and a design was specifically targeted which could both work in a lab environment and be easily extended to a poolside or dockside test area. As shown in Figure 3.27, the rig consists of a hanging cable from which a scale and CALIPSO is suspended, all weighed down by an anchor mass.

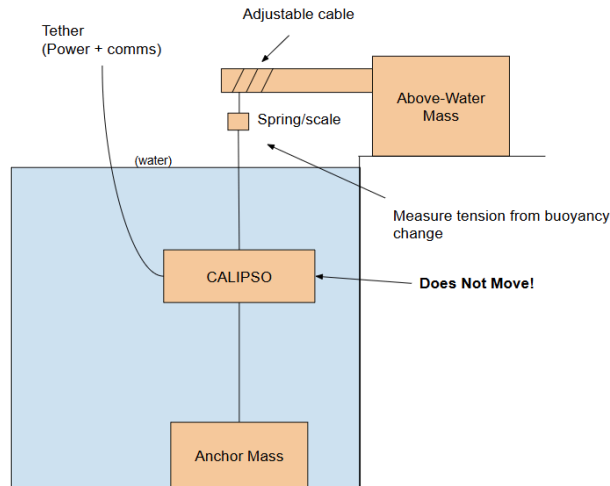


Figure 3.27: Testing Setup Concept

3.7.2 Physical Test Setup

The system was suspended in tension and attached to a hanging scale, and the team used adjustable zip ties as a straightforward and repeatable tensioning tool. These are specifically a zip tie style that can be tightened and undone repeatedly. A large testing bin was filled with water for the system to be submerged in. With a large enough volume, the weight of the water became significant, leading the plastic bin to bow outwards under the weight. Slight support was introduced by adding solid mass to the container, here bricks, to displace water and reduce the outwards pushing force from water. Duct tape, similarly, helped resist this outwards push. A more rigid container would be a worthwhile future improvement, such as a large size Pelican case.

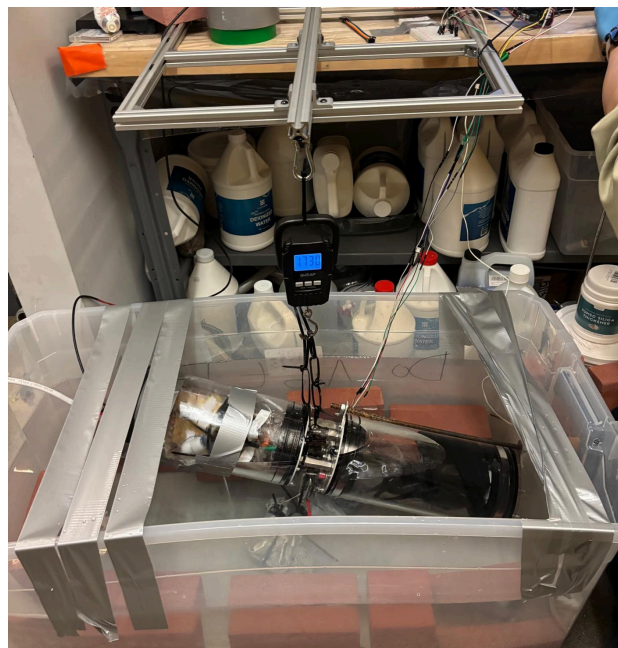


Figure 3.28: Real Testing Setup Implemented

4. Results

4.1 Water & Aluminum Reaction Rate Characterization

To get a better understanding of the reaction rate and yields of our aluminum fuel paste we conducted multiple tests. The first test that resulted in notable reaction information was when we reacted to the aluminum fuel with tap water inside the bladder. This reaction resulted in a very slight temperature change in the bladder (22 C to 29 C), but had no visible gas inflation. We believe that only a small fraction of the fuel reacted due to mixing and pumping failures. However, when flushing the fuel bag for this test, a notable heat increase and pressure build up did occur. Unfortunately, since this occurred during our clean up process, no data was taken. The reaction started about 20 minutes after water was introduced, which showed a very delayed reaction. We attribute the delayed reaction to the time the water took to permeate the oil the fuel was suspended in.

Next, to increase reaction rate and yields, we tested the reaction between aluminum fuel and distilled water, with caffeine as a catalyst. In testing done with aluminum pellets, distilled water has shown better yield than tap water, with caffeine acting as a catalyst for the reaction [22]. We set this system up to a beaker apparatus to test the hydrogen yield values. Unfortunately, due to ruptures in our bladder, hydrogen was not able to accumulate. However, we could see fluctuations in the gas level in the outlet tube (we believe this is due to the gas level increasing until pressure built up enough to leak out the rupture) which showed us that hydrogen was being produced.

With the rest of the aluminum fuel paste we had, which was about 3 mL of paste (~1.5 grams of aluminum due to excess oil and settling of aluminum), we hoped to get a volume yield value using a test tube setup. This reaction was quick to start and produced over a liter of hydrogen gas before the reaction was finished. The beaker we were measuring hydrogen volume with was only 1 L and that was filled about half way through the reaction and extrapolating that would result in about 2 L of hydrogen. From testing the reaction yield of an aluminum pellet (0.302 grams of aluminum at 100% yield) results in about 400mL of hydrogen gas. This means that the fuel we were testing had close to 100% yield (5 times the amount of aluminum resulted in about 5 times the amount of volume). To note, the reaction required a significant amount of mixing to ensure all the fuel was reacted properly and was very exothermic.

This testing with the aluminum fuel paste did not result in too much quantitative data due to issues with fuel delivery, mixing, and the experimental setup as well as limitations in fuel production due to the danger of aluminum powder. However, there are good qualitative takeaways from this experimental testing. These are the fact that oil delays the start of the reaction, mixing will not occur spontaneously in the bladder (but rather a reaction chamber or tube should be more utilized), that the aluminum fuel yield is on par with pure aluminum, and that the reaction rate is much better with DI water and caffeine.

4.2 Testing Control System

Prior to any in-water trials, the CALIPSO control logic and actuation hardware were subjected to an exhaustive dry-run campaign. All pumps, peristaltic aluminum paste, peristaltic water, diaphragm gas vent, and diaphragm hull intake, were exercised through their full ranges of PWM duty cycles to verify dispense timing accuracy and repeatability. Using the Commands Dashboard GUI, we issued ASCEND, DESCEND, ABORT ASCEND, ABORT DESCEND commands in isolation and in sequence, confirming that the ESP32 logic and pump motors responded as expected. During these dry tests, we monitored the tubing, sensors, pumps, and enclosures to ensure no unintended actuation or backflow occurred.

Building on the dry-run results, we conducted multiple in-water trials in a tank. The runs were often used one at a time to test a different part or function of CALIPSO. The runs began with filling up the hull and then testing the reactant dispensing, the hull intake, and the bladder gas venting. Real-time feedback was displayed on the Sensors Dashboards, which allowed us to monitor temperature and pressure in case abort commands were needed. The challenges encountered were mostly regarding the pumping of aluminum paste. The duty cycle of the aluminum pump had to be updated and the reactant mixing process was changed to reflect the paste's viscosity characteristics.

This testing sequence demonstrated a comprehensive validation of CALIPSO's control system. With the dry runs, the team isolated and tuned actuation timing for all pumps, while the in-water trials verified the real-world performance of the system under varying conditions. The tests confirmed the performance and efficiency of our open-loop control strategy, GUI commands, sensor feedback, and abort mechanisms.

4.3 Overall System Testing Outcome

Final system testing results for this proof of concept configuration revealed a number of insights into the system. Testing was conducted first with the aluminum fuel paste and distilled water within the test setup multiple times, but the fuel paste did not have a fast enough reaction time to cause appreciable temperature gain and the resultant water vaporization resulting in lower gas volume. This was compounded by an unexpectedly low yield for the paste which yielded hydrogen gas volumes far below what was required to both inflate the bladder and thus cause a change in buoyancy. Following these testing issues the system was tested on a lab bench in a dry environment with the aluminum fuel paste dispensed directly into the reaction bladder with a syringer and distilled water added. This configuration was to remove the potential oxidation time associated with loading the fuel into the bag, assembling the configuration, and filling the hull. However, even in this modified and simpler setup the reaction still did not produce enough gas yield to significantly inflate the bladder within the time required and instead took around 10 minutes to inflate the bladder below the threshold that was needed. It was determined through subsequent lab testing that the current fuel only reacts well if it is allowed to mix with distilled water and rest for a significant time within a confined area. These reaction dynamics are not ideally suited to the current mixing setup targeting laminar flow mixing before

depositing the reactants into the bladder. Lab testing ultimately determined that the fuel and water were too dispersed within the bladder to deliver the yield required, and instead the water would need to penetrate through the oil by sitting within a confined area instead of the vigorous mixing regime this design was optimized for. Although this result is not desirable the manufacturing of the fuel itself was outside the scope for this design cycle and as such a separate fuel testing setup was pursued to validate the use of solid fuel to generate buoyancy and prove out the fundamentals of an aluminum fuel buoyancy engine.

To prove these fundamentals solid fuel aluminum pellets were instead used as they have a higher yield and quicker reaction time allowing for the rapid buoyancy generation we are targeting. The test setup was reset and five aluminum fuel pellets were added to the reaction bladder, equalling around 0.5 mL of aluminum fuel. Of note, during this test the reaction bladders were switched due to splitting of the 3D printed reaction bladders around the interfaces which prevented them from trapping produced gas inside. This was caused due to the natural weakness between layers introduced in the process and the irregularity of the printed interfaces which lead to high stress concentrations and crack propagation once barb fittings were attached. Instead a latex blood pressure cuff was used which held the same volume as the 3D printed bladder. The test setup was assembled and the enclosure flooded before distilled water was pumped into the bladder. Once the water entered the bladder the reaction initiated in seconds and after 10 seconds had passed from initiation the bladder was fully inflated.

Ultimately based on the scale readouts the reaction generated approximately 0.975 kg of buoyant force which equates to 975 mL of gas generation. The reaction had purposely been increased when the decision was made to switch to the solid fuel pellets, to prove the concepts of the test configuration which explains the larger quantities of gas produced. Following the production of the gas, the bladder was fully vented using the gas pump and then the sea water pump successfully forced the gasses out of the check valve and refilled the flooded hull returning to a force reading within 0.1 kg of the original reading when the hull was fully flooded. The discrepancy in measurements between the initial descent cycle and the second descent can be explained by the orientation of the check valve which did not allow for complete gas venting. To solve this is a later design the check valve would simply have to be moved and another one installed so that one is on each end of the flooded hull, allowing for complete gas venting regardless of angle of attack of the buoyancy engine.

Temperature and pressure were recorded throughout this bladder inflation as shown below in Figure 4.2 and 4.3. Peak overpressure during the reaction was over 6 psi which is three times the cracking pressure of the check valve. This overpressure happened due to the check valve not being able to properly vent the pressure produced rapidly enough to equalize the internal pressure. However this overpressure rapidly dissipated over time before equalizing to the external pressure. The temperature peaked around 80 C due to the speed of the reaction initiation and propagation. The testing data from the aluminum fuel paste reaction is also shown in Figure 4.1 which notably lacks large temperature peaks due to the issues discussed prior with reaction initiation. The aluminum fuel paste testing took place on the benchtop within the reaction

bladder which is the reason for the absence of the pressure data. This decision was made based on the issues in reacting the paste within the chamber.

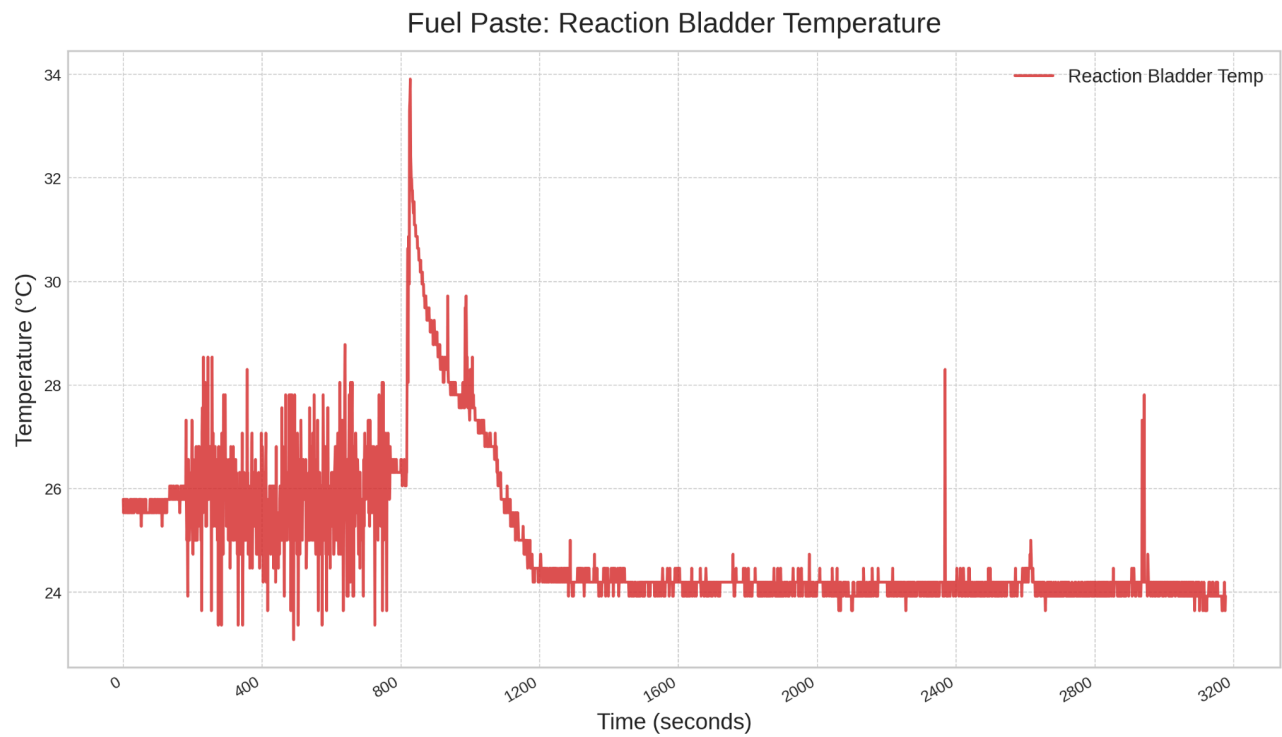


Figure 4.1: Graph of the reaction temperature over time for the aluminum fuel paste reaction bladder benchtop testing.

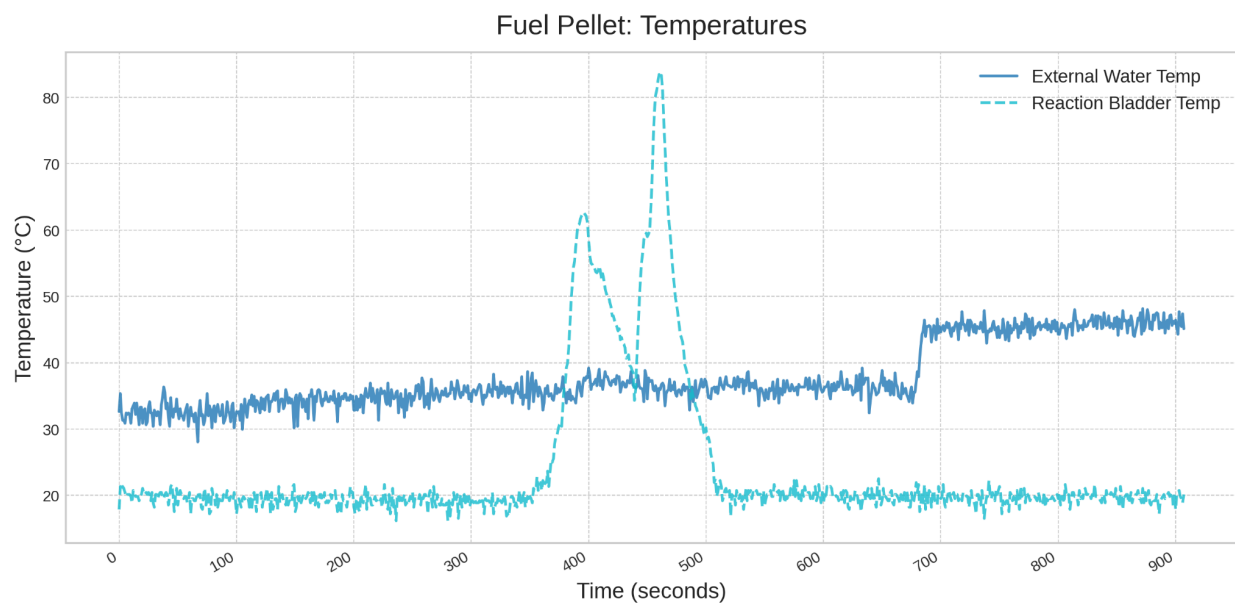


Figure 4.2: Graph of the temperature over time of the reaction bladder and external water during the fuel pellet experiments with the test configuration.

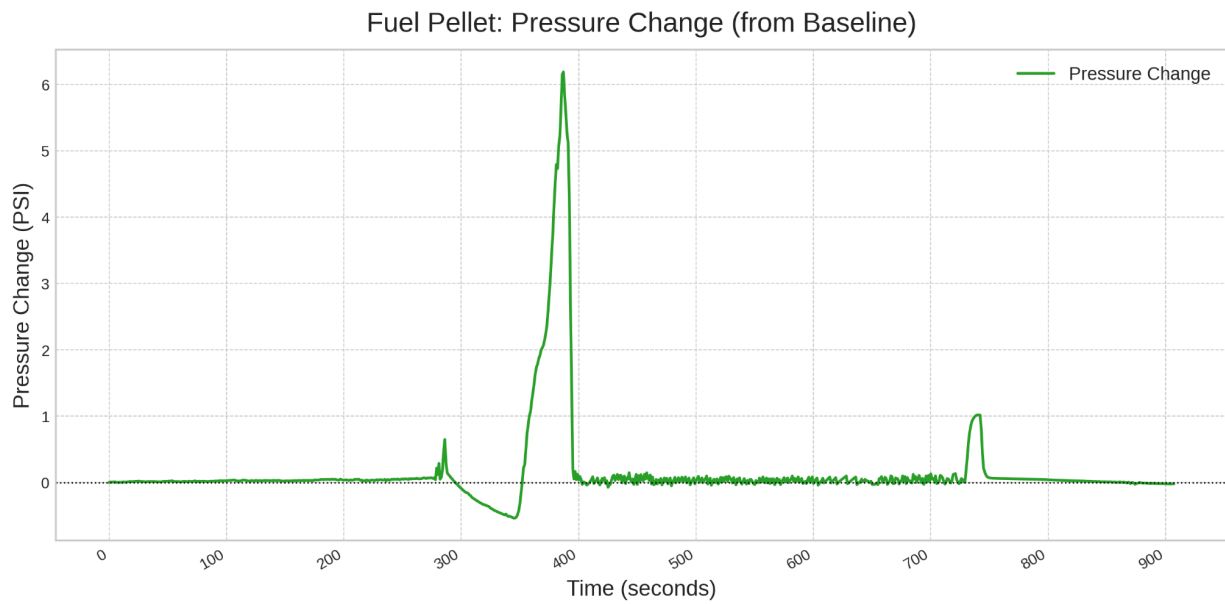


Figure 4.3: Graph of the pressure change over time during the fuel pellet experiments with the test configuration.

5. Bill of Materials

Table 5.1: A summarized bill of materials cost for the test configuration constructed during this design cycle.

Subsystem	Part Description	Total Price (\$)
Enclosure	Enclosure Hardware	\$1,870.00
EE/Controls	Control System/Electronics	\$484.00
Test Setup	Hardware for Test Frame	\$97.00
Reactant System	Reaction Bladder	\$300.00
Reactant System	Tubing and Hardware	\$137.00
Reactant System	Reactant Production	\$33.00
Reactant System	Peristaltic Pump	\$26.00
Reactant System	Diaphragm Pump	\$24.00
Reactant System	Blood Pressure Cuff Bladder	\$15.98

6. Scaling Up / Future Work

For the scope of this project, the functional requirements were significantly decreased from PUMPY (a full scale, functional buoyancy engine that was rated for depths of up to 500 m below the surface). The goal of CALIPSO was instead to prove that a buoyancy engine fueled by aluminum fuel is a possibility. CALIPSO showed that the aluminum fuel was pumpable, that the aluminum and water reaction can generate buoyancy, and that a full dive cycle is possible (flooding the hull, producing gas from the reaction, venting the bladder, and releasing gas through the check valve). All of these accomplishments can be leveraged to prove that a full scale buoyancy engine is possible by replacing hardware, introducing ammonia, and increasing fuel requirements.

6.1 Hardware

Although the main design of CALIPSO can be used as a scaled up buoyancy engine, a couple key things would need to change. In order to ensure that the hardware can be scaled up to depths suitable for a buoyancy engine, the major concern is the pressures at depth (500m $\rightarrow$ 50 atm). In order to ensure that there are no significant pressure differentials across the system, CALIPSO would need to be filled with incompressible liquids. Since the flooded chamber is already flooded, the only change would be the electronics container will need to be filled with an incompressible, dielectric oil. If all the containers are flooded, there will be no significant pressure differences that the pumps will be pumping across. Therefore, the pumpheads we rated are suitable, the pumps just need to be rated for depths so that they do not contain any air and can work fully submerged which are common pumps in submersibles. Furthermore, to ensure no gas/fuel leakage from the bladder/tubing, the connectors and fittings should be reinforced and sealed with epoxy, hard tubing should be used, and the cable glands should be replaced with custom connectors for both the tubing and the electronics cord. Finally, putting this system into one hull will allow for rearranging of systems to shrink the length and ensure it can fit as a subsystem of a submersible. The hardware for all these changes would need to be sourced, but is a very feasible task.

6.2 Temperature and Pressure

The other main concern is the cold temperature and high pressure at depth. Temperature and pressure concerns come into play since the gases produced by the reactions are compressible and will have a much smaller volume yield in high pressure environments as well as the fact that there will be significantly more heat transfer out of the system if the ambient temperature is 0 °C, which it is at 500 m. These concerns can be overcome by a combination of two things. First, by using ammonia. Since ammonia has a much lower enthalpy of vaporization than water [4], it will require much less energy to become gaseous. Furthermore ammonia is a catalyst to help accelerate the reaction [22], which means there is a shorter time frame for heat transfer to occur (more heat can go to gas expansion). Second by increasing the fuel amount. The amount of fuel needed to fully inflate the bladder at 1.5 atm is 0.21 mL. Since the pressure is linearly related to

the amount of fuel, if we scale up the pressure to 50 atm, the order of magnitude of fuel size will scale up about 30 fold which would mean the fuel requirement becomes about 7-10mL of paste required. Given the fuel bag was about 200mL, that would mean at least 20 cycles with room to scale as needed. Therefore, despite some changes to design, hardware, reaction, and the system being necessary when scaling up, CALIPSO proves that a fully functional aluminum powered buoyancy engine is possible.

Furthermore, despite the possibility to leverage a full scale buoyancy engine from CALIPSO, there is a much more practical and necessary use for CALIPSO. That is as a means to test, experiment, and iterate. There is not much information on the aluminum fuel and its reaction. There is much to learn about different mixing systems, different pumps, fuel viscosities, and even reaction rates. Therefore, due to its modular and flexible design, CALIPSO is a great system to use for these tests. Before scaling and trying to create a fully fledged out buoyancy engine, much more research and testing with CALIPSO is needed to be done to bring much needed understanding to this type of engine.

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